Chapter-03:
Cell Division and Patterns of Inheritance

Ref book: Biology for Engineers - Arthur T. Johnson [2nd edition]
Biology for Engineers – G. K. Suraishkumar

BIO-3105, Biology for Engineers
Lecture-08; 15/03/2025
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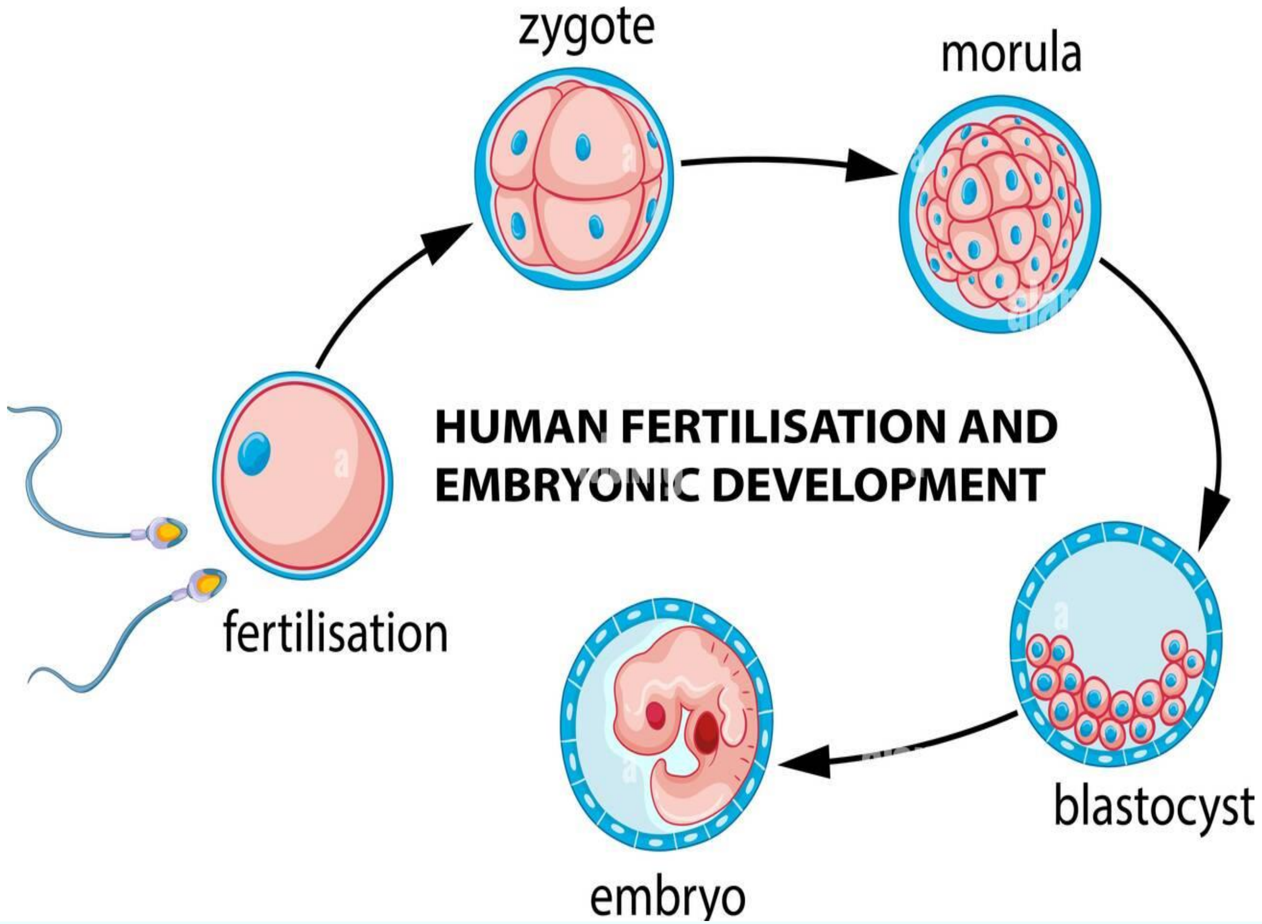
From Zygote to Adult and Death!



Cell Division:

A. In Prokaryotes: cell division is **simply to divide the contents.**

B. In Eukaryotes: the division is more complex it occurs in two stages:

1. **Mitosis:** division of the **nucleus**
2. **Cytokinesis:** division of the **cytoplasm**

Unicellular Vs Multicellular

- **Unicellular:** organisms use asexual reproduction methods such as fission, budding, etc. Cell division basically consists of two processes karyokinesis and cytokinesis.
- This type of asexual reproduction producing **two identical daughter** cells from one parent cell.

- **Multicellular:** organisms use mitosis and cytokinesis to generate **new cells to the organism** for growth and developments.

Chromosomes>DNA

Chromosomes are composed of DNA and proteins (histones) and carry the genetic information in eukaryotic cells.

Each species of organism has a specific number of chromosomes.
Drosophila *melanogaster* has **8** chromosomes
Humans have **46** chromosomes(23 Pairs, 22 pairs autosomes+1sex chromosomes)
Dogs have **78** pair of chromosomes.

Chromosomes **are not visible** except during cell division.
Replication (copying) of the DNA occurred before cell division, therefore every chromosome is actually two identical sister chromatids.
Each pair of chromatids is connected to each other at an area called the **centromere,** usually located near the center of the chromosome.

Cell Cycle and Cell Division

Cell division is just one of several stages that a cell goes through during its lifetime.

The cell cycle is a repeating series of events that include growth, DNA synthesis, and cell division.

Where does the cell division occur?
Mitosis is a process of nuclear division in **eukaryotic cells** that occurs when a parent cell divides to produce two identical daughter cells. During cell division, mitosis refers specifically to the separation of the duplicated genetic material carried in the nucleus.

Why Cells Divide?

- Repair of damaged tissue.
- Growth and development of an organ.
- Replace old and dying cells.

- **How do cells grow and divide?**
- When cells divide and grow they do this very precisely so that the new cells are exactly the same as the old ones. **Each cell makes copies of all its genes. Then each cell splits into 2 with one set of genes in each new cell**. During the process, there are lots of checks to make sure that everything has copied correctly.

Why do cells stop dividing?

Aging mammalian cells can stop dividing and enter senescence **if they are damaged or have defective telomeres**. Senescence protects against tumor formation, and tumor suppressor genes include some that regulate cell division and lead to senescence.

Which cells do not divide?

Nerve cell does not divide because they do not have **centrioles**, so they cannot undergo mitosis and divide to form new cells.

What part of the cell divides first?

The nucleus, when a eukaryotic cell divides, **the nucleus** divides first in the process of mitosis

Centrioles – Centrioles are present in the animal cells. They duplicate during the cell division and move to the opposite pole and play a role in anchoring spindle fibres, aligning and separating the chromosomes.

Types of cell division?

- **Mitosis and**
- **Meiosis** are the two main types of cell division

What is the main difference between mitosis and meiosis?

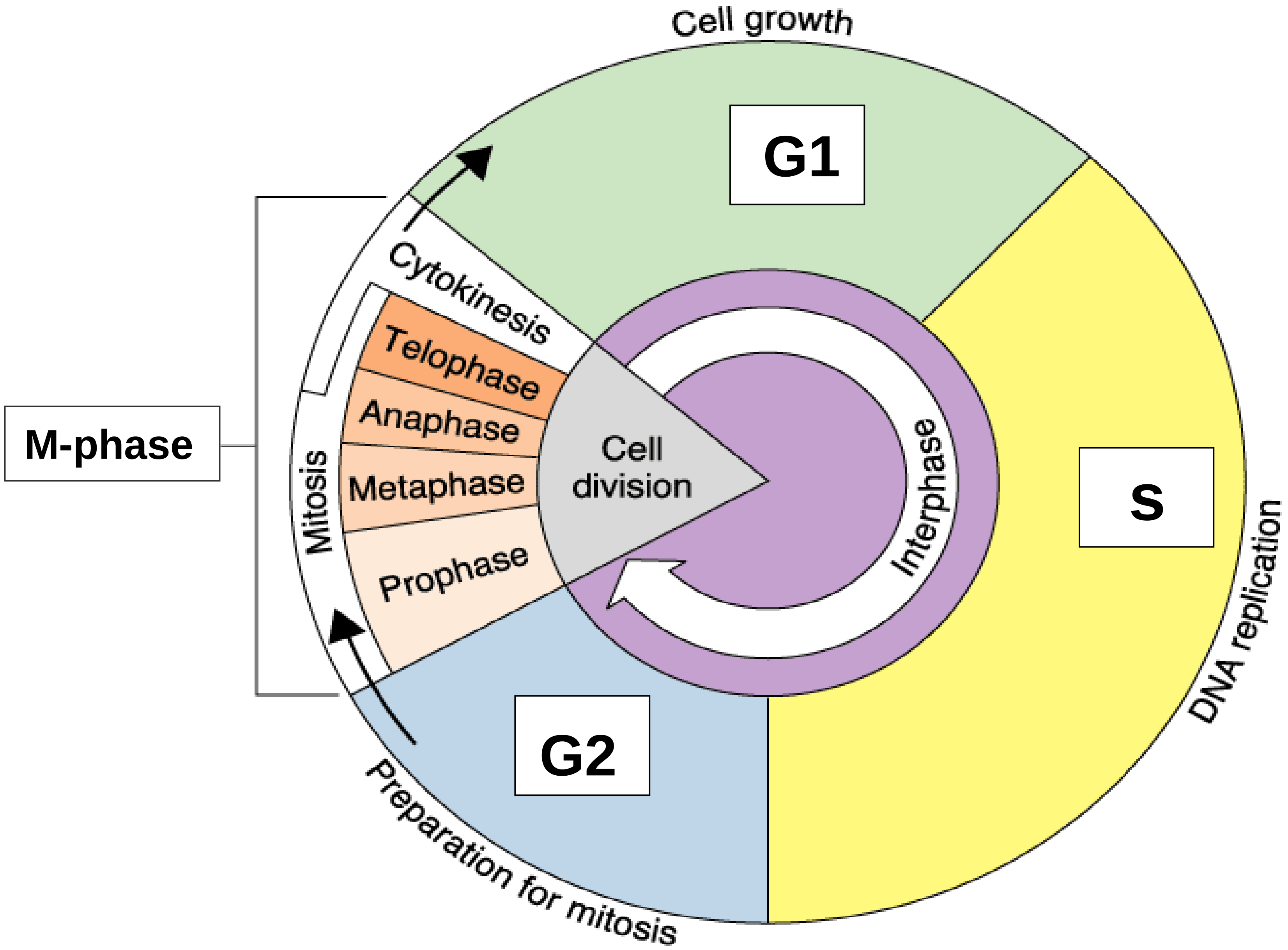
Cells divide and reproduce in two ways, mitosis and meiosis.

- **Mitosis results in two identical daughter cells,**
- **Meiosis results in four gamet cells.**

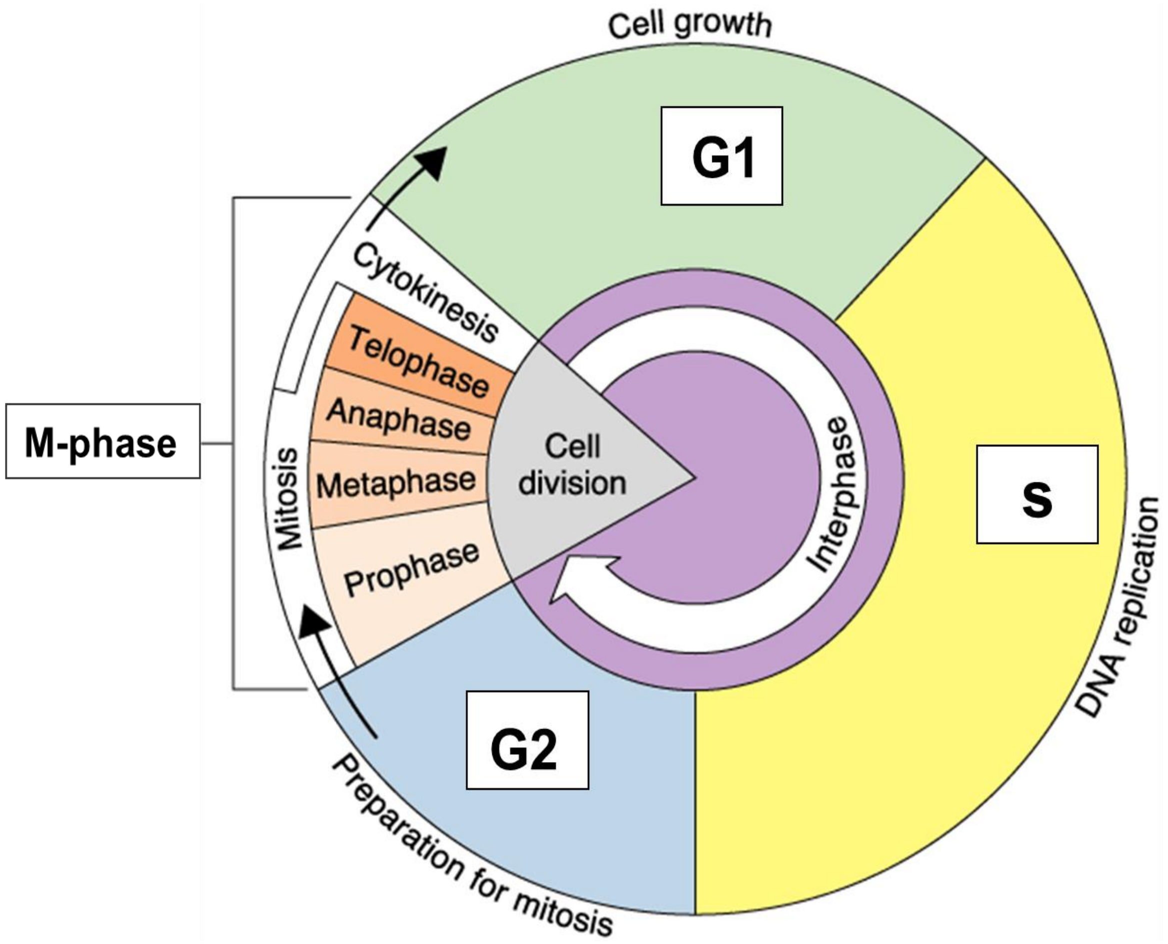
MITOSIS

They are also genetically identical to the parental cell. Mitosis has five different stages: **interphase, prophase, metaphase, anaphase and telophase**. The process of cell division is only complete after cytokinesis, which takes place during anaphase and telophase

Events of the Cell Cycle



Interphase:



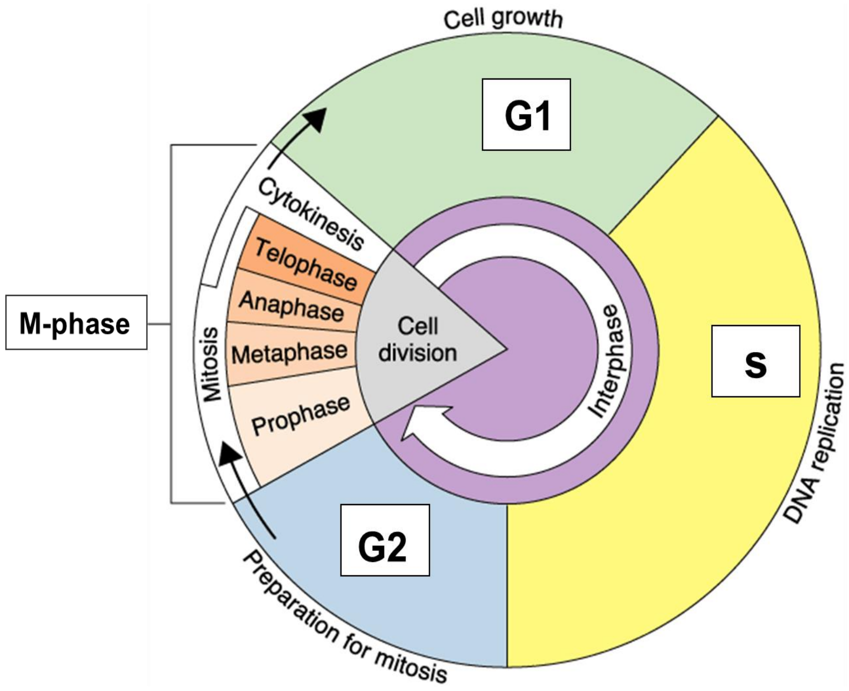
What happens in the 3 phases of interphase?

There are three stages of interphase: G_1 (first gap), S (synthesis of new DNA), and G_2 (second gap).

Interphase is defined by three stages: **the first gap phase (G1), synthesis (S) phase, and the second gap (G2) phase**.

Cells spend most of their lives in interphase, specifically in the S phase where genetic material must be copied. **The cell grows and carries out biochemical functions, such as protein synthesis, in the G_1 phase.**
At the end of interphase comes the mitotic phase, which is made up of mitosis and cytokinesis and leads to the formation of two daughter cells.

Interphase



- Stages of Interphase
- During interphase, the cell undergoes normal growth processes while also preparing for cell division. It is the longest phase of the cell cycle, cell spends approximately 90% of its time in this phase. In order for a cell to move from interphase into the mitotic phase, many internal and external conditions must be met.
- The three stages of interphase are called G1, S, and G2.
- **G1 Phase (First Gap)**
- The first stage of interphase is called the G1 phase (first gap) where the cell is quite active at the biochemical level. The cell is accumulating the building blocks of chromosomal DNA and the associated proteins as well as accumulating sufficient energy reserves to complete the task of replicating each chromosome in the nucleus.
- **S Phase (Synthesis of DNA)**
- In the S phase, DNA replication results in the formation of identical pairs of DNA molecules—sister chromatids—that are firmly attached to the centromeric region.
- **G2 Phase (Second Gap)**
- In the G2 phase, the cell replenishes its energy stores and synthesizes proteins necessary for chromosome manipulation. Some cell organelles are duplicated, and the cytoskeleton is dismantled to provide resources for the mitotic phase.

MITOSIS:

- The mitotic phase can be sub-divided into four phases called **Prophase, Metaphase, Anaphase and Telophase (PMAT).**
- Mitosis is strictly **nuclear** division.
- Mitosis is followed by cytoplasmic division, called **cytokinesis** to complete cell division

What are the 4 phases of mitosis and what happens in each steps?

1) **Prophase:** chromatin into chromosomes, the nuclear envelope break down, chromosomes attach to spindle fibers by their centromeres

2) **Metaphase:** chromosomes line up along the metaphase plate (centre of the cell)

3) **Anaphase:** sister chromatids are pulled to opposite poles of the cell

4) **Telophase:** Separation by nuclear envelope.

DNA

Most DNA is located in the cell nucleus (where it is called nuclear DNA), but a small amount of DNA can also be found in the mitochondria (where it is called mitochondrial DNA or mtDNA). Mitochondria are structures within cells that convert the energy from food into a form that cells can use.

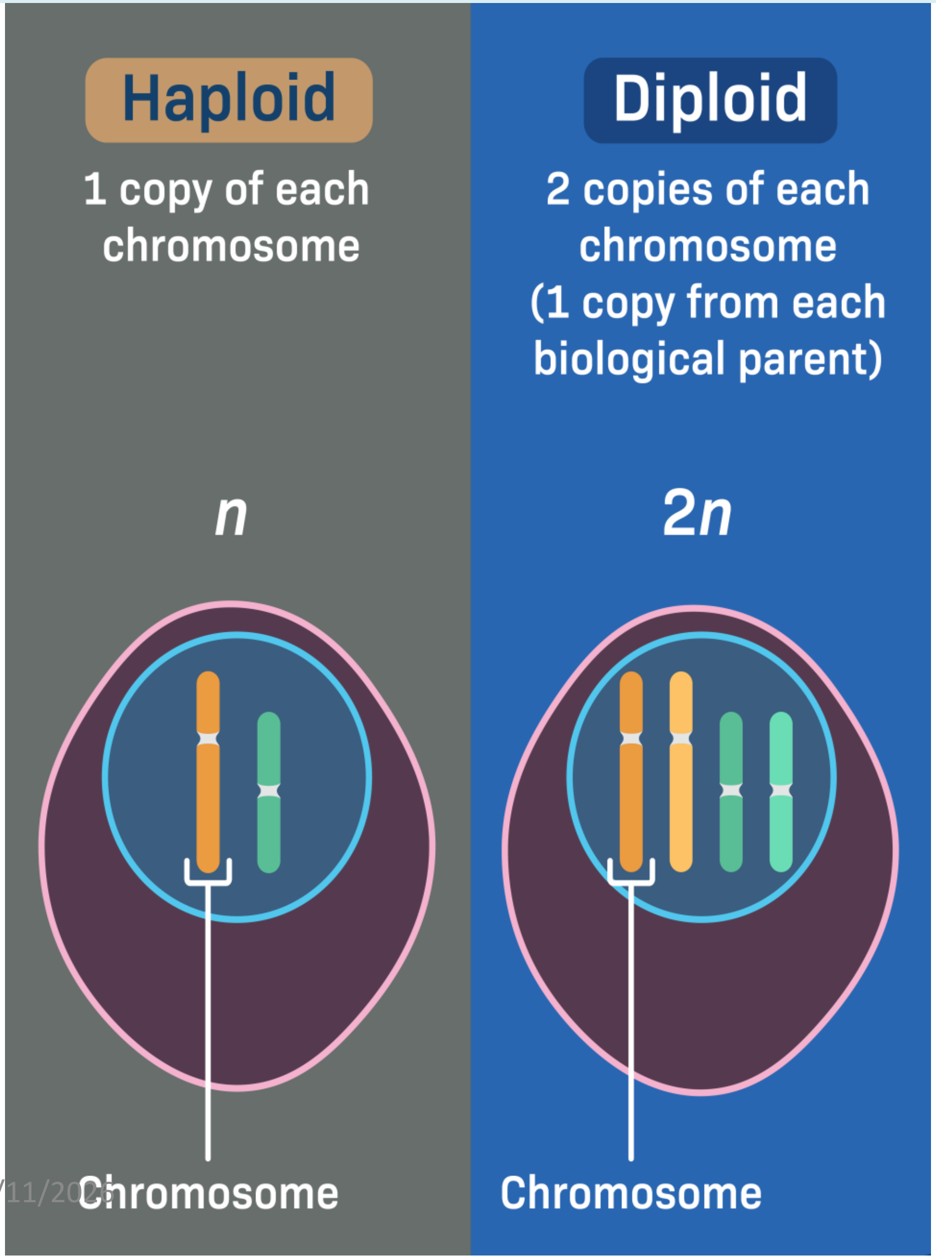
What is haploid and diploid?

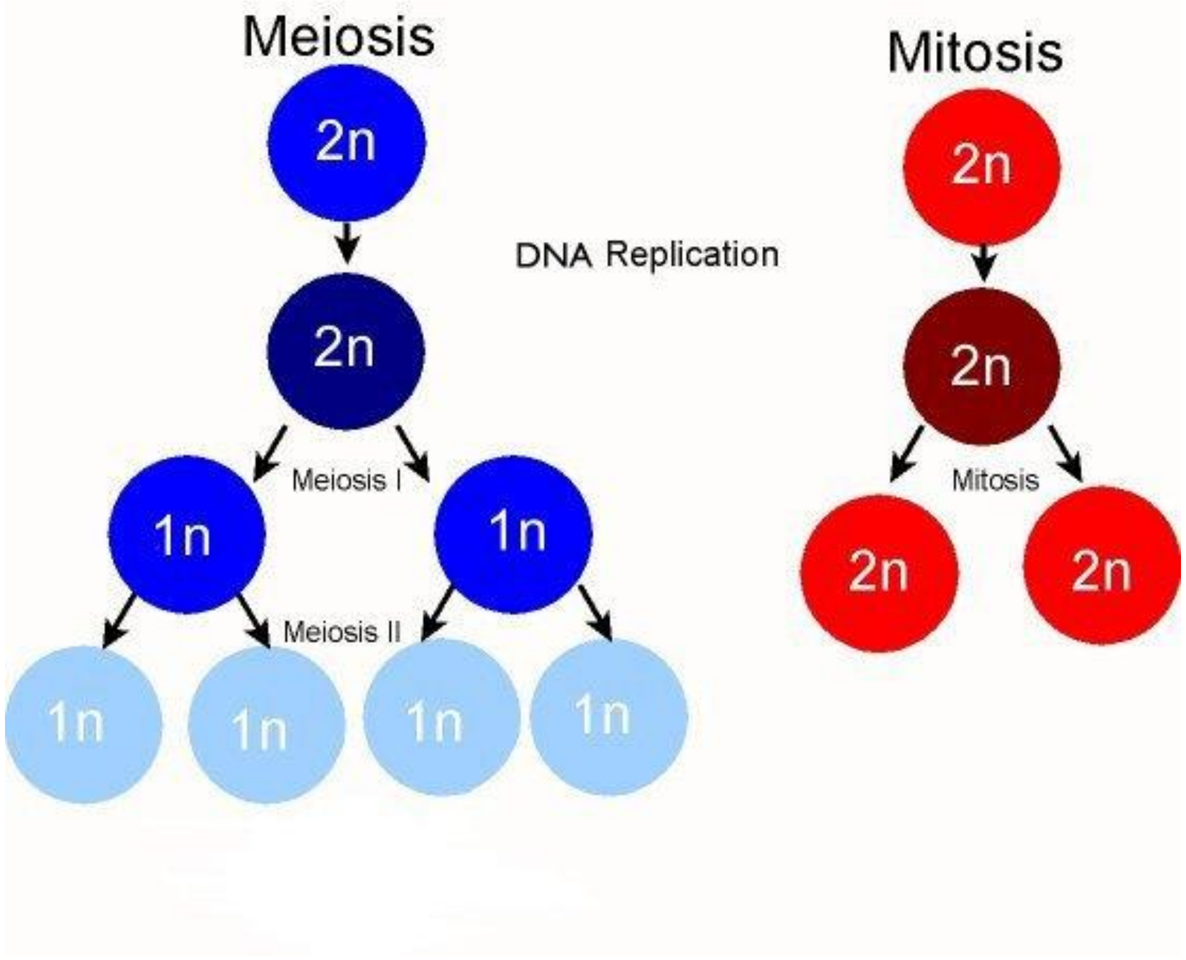
Haploid refers to the presence of a single set of chromosomes in an organism's cells. Sexually reproducing organisms are diploid (having two sets of chromosomes, one from each parent). In humans, only the egg and sperm cells are haploid.

HOW GENES are NAMED?

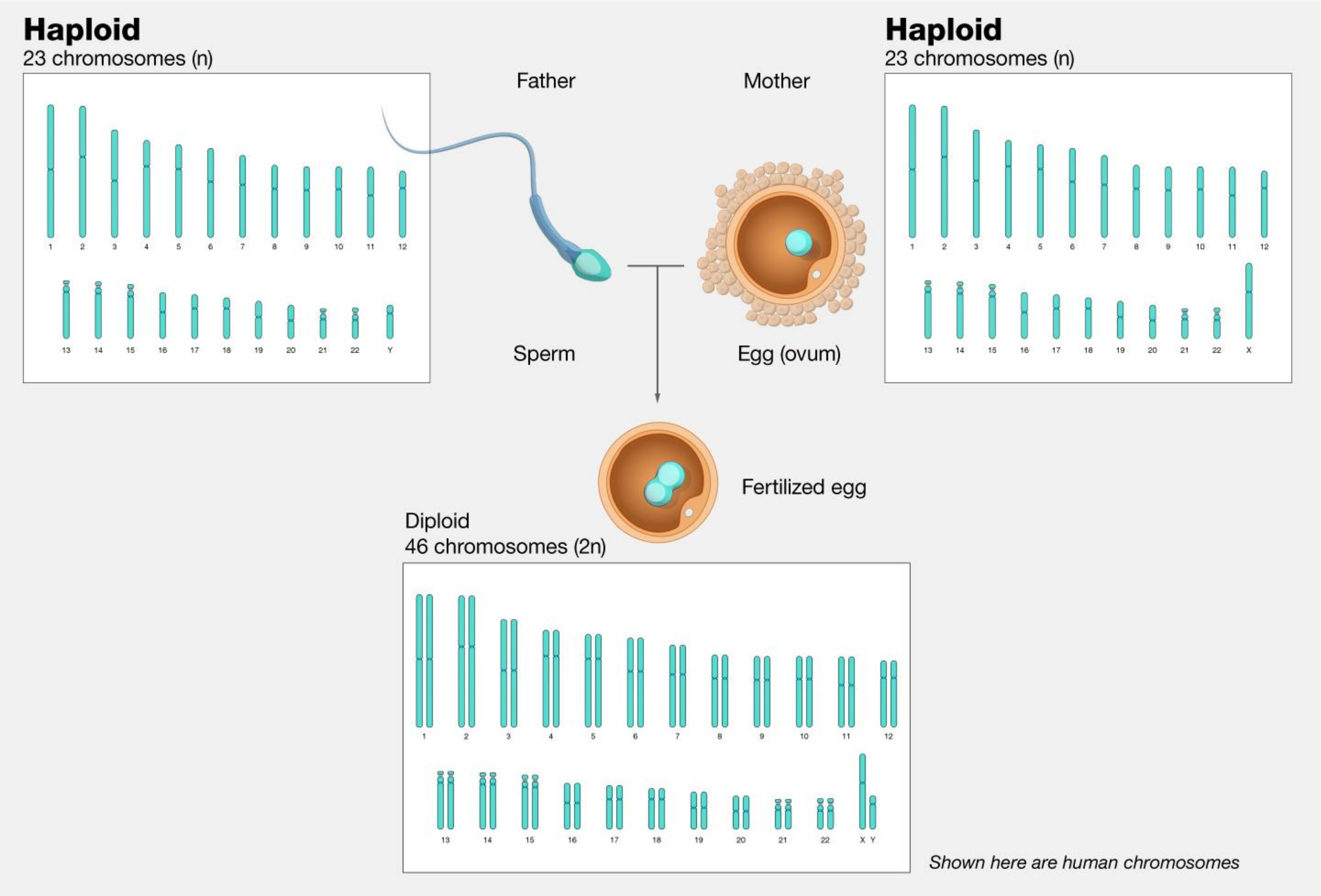
When a scientist discovers a new human gene, he or she submits a proposed name to the Human Gene Nomenclature Committee (HGNC), an international panel of researchers with exclusive authority over this area. Guidelines were established in 1979 by the HGNC and have been updated periodically

HAPLOID/DIPLOID





HAPLOID/DIPLOID



HAPLOID/DIPLOID

Is meiosis haploid or diploid?

Meiosis is the production of four genetically diverse **haploid daughter cells** from one diploid parent cell. Meiosis can only occur in eukaryotic organisms.

How is Meiosis I Different from Meiosis II?

Meiosis I	Meiosis II
Starts as diploid; Ends as haploid	Starts as haploid; Ends as haploid

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Introduction: Meiosis

(a) (b) (c)

Sexual reproduction, specifically meiosis and fertilization, introduces variation into offspring that may account for the evolutionary success of sexual reproduction. The vast majority of eukaryotic organisms, both multicellular and unicellular, can or must employ some form of meiosis and fertilization to reproduce.

Meiosis employs many of the same mechanisms as mitosis. *However, the starting nucleus is always diploid and the nuclei that result at the end of a meiotic cell division are haploid. To achieve this reduction in chromosome number, meiosis consists of one round of chromosome duplication and two rounds of nuclear division.*

Because the events that occur during each of the division stages are analogous to the events of mitosis, the same stage names are assigned. However, because there are two rounds of division, the major process and the stages are designated with a "I" or a "II." Thus, **meiosis I** is the first round of meiotic division and consists of prophase I, prometaphase I, and so on. **Meiosis II**, in which the second round of meiotic division takes place, includes prophase II, prometaphase II, and so on.

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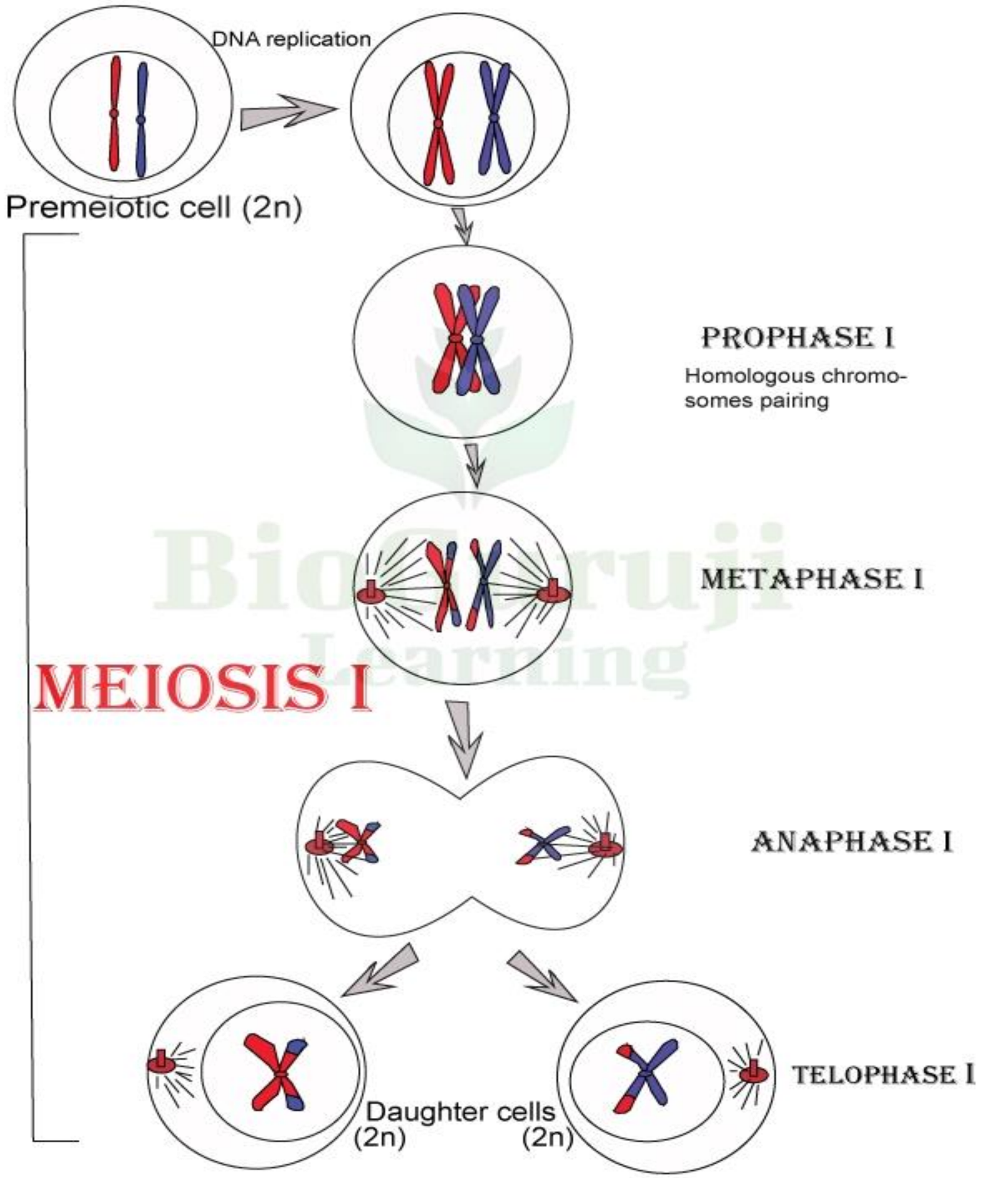
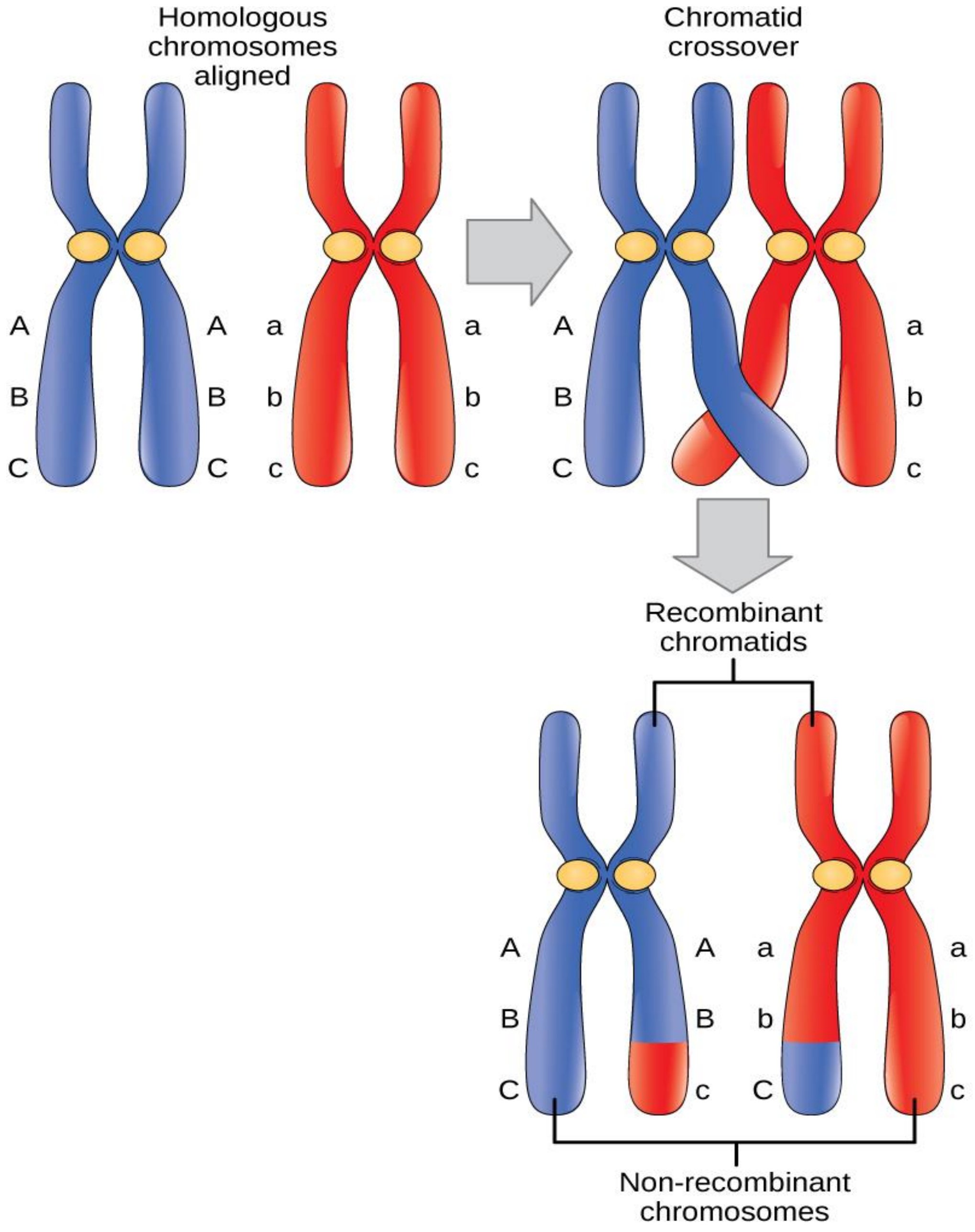
Meiosis I

- Meiosis is preceded by an interphase consisting of the G_1, S, and G_2 phases, which are nearly identical to the phases preceding mitosis. The G_1 phase, which is also called the first gap phase, is the first phase of the interphase and is focused on cell growth. The S phase is the second phase of interphase, during which the DNA of the chromosomes is replicated. Finally, the G_2 phase, also called the second gap phase, is the third and final phase of interphase; in this phase, the cell undergoes the final preparations for meiosis.
- During DNA duplication in the S phase, each chromosome is replicated to produce two identical copies, called sister chromatids, that are held together at the centromere. The centrosomes, which are the structures that organize the microtubules of the meiotic spindle, also replicate. This prepares the cell to enter prophase I, the first meiotic phase.



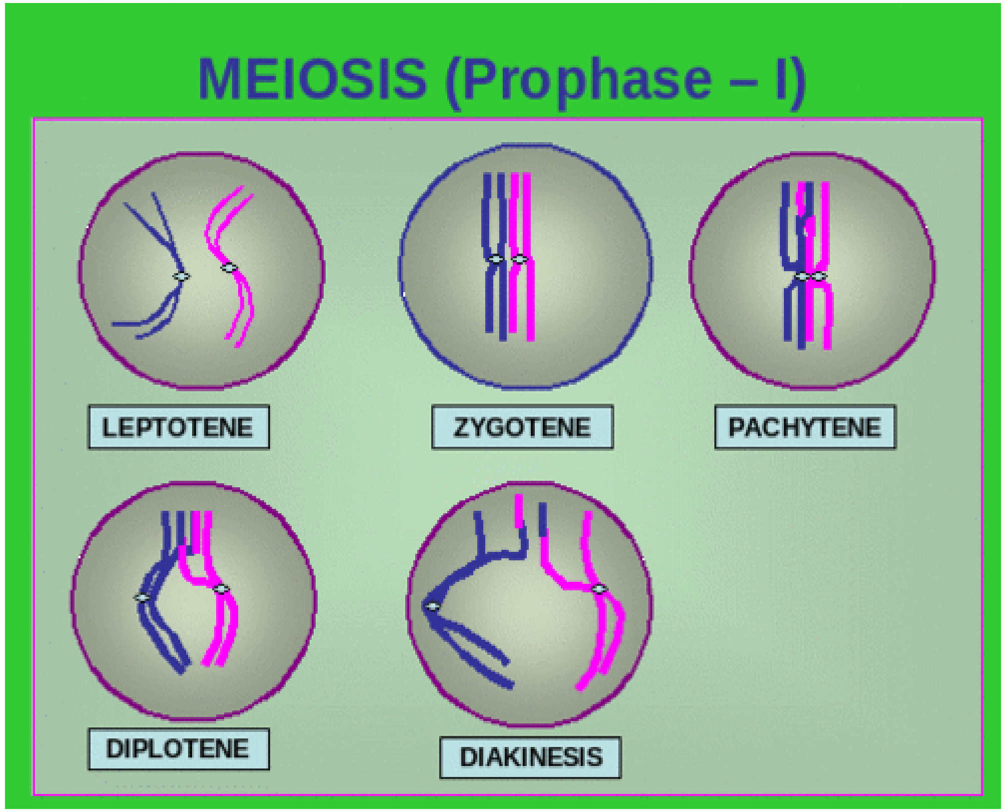
Meiosis I: Prophase I

- As the nuclear envelope begins to break down, the proteins associated with homologous chromosomes bring the pair close to each other. (Recall that, in mitosis, homologous chromosomes do not pair together. In mitosis, homologous chromosomes line up end-to-end so that when they divide, each daughter cell receives a sister chromatid from both members of the homologous pair.) The tight pairing of the homologous chromosomes is called **synapsis**. In synapsis, the genes on the chromatids of the homologous chromosomes are aligned precisely with each other (Figure 1). The synaptonemal complex supports the exchange of chromosomal segments between non-sister homologous chromatids, a process called **crossing over**. Crossing over occurs at **chaiasmata** (singular = chiasma), the point of contact between non-sister chromosomes of a homologous pair .
- At the end of prophase I, the pairs are held together only at the chiasmata and are called **tetrads** because the four sister chromatids of each pair of homologous chromosomes are now visible.



Meiosis I: Prophase I

- The crossover events are the first source of genetic variation in the nuclei produced by meiosis. A single crossover event between homologous non-sister chromatids leads to a reciprocal exchange of equivalent DNA between a maternal chromosome and a paternal chromosome. Now, when that sister chromatid is moved into a gamete cell it will carry some DNA from one parent of the individual and some DNA from the other parent. Multiple crossovers in an arm of the chromosome have the same effect, exchanging segments of DNA to create recombinant chromosomes.
- A second event in Prophase I is the attachment of the spindle fiber microtubules to the kinetochore proteins at the centromeres. At the end of prometaphase I, each tetrad is attached to microtubules from both poles, with one homologous chromosome facing each pole. The homologous chromosomes are still held together at chiasmata.
- In addition, the nuclear membrane has broken down entirely.



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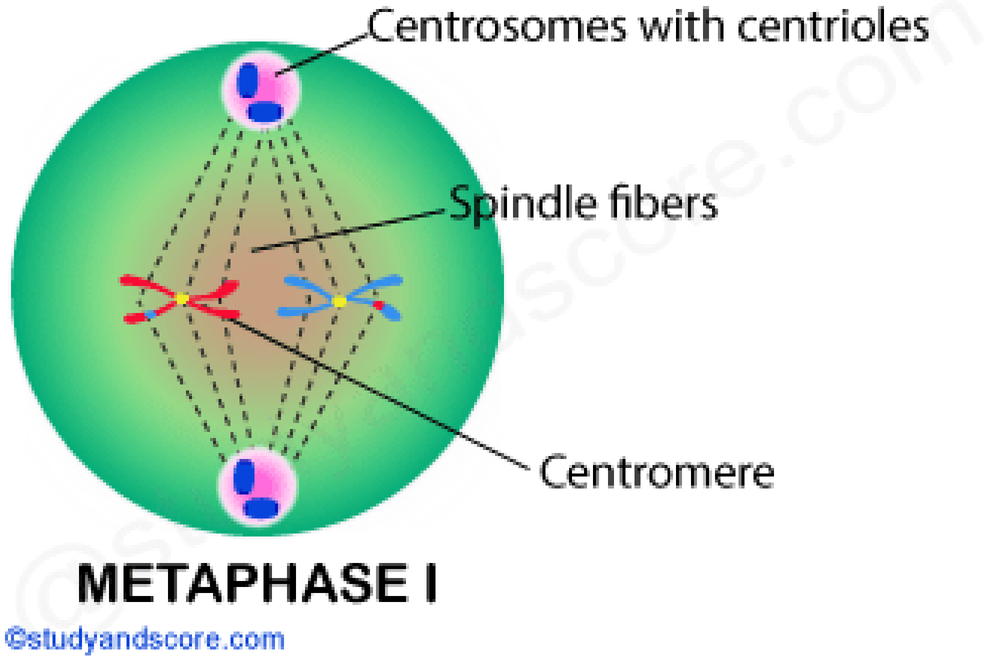
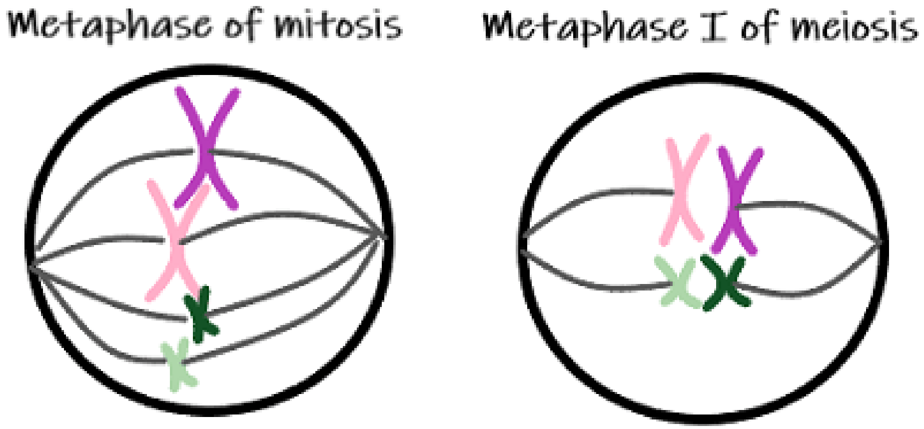
Meiosis I: Metaphase I

- During metaphase I, the homologous chromosomes are arranged in the centre of the cell with the kinetochores facing opposite poles. The homologous pairs orient themselves randomly at the equator. Recall that homologous chromosomes are not identical. They contain slight differences in their genetic information, causing each gamete to have a unique genetic makeup. **This randomness is the physical basis for the creation of the second form of genetic variation in offspring**. The number of variations is dependent on the number of chromosomes making up a set. There are two possibilities for orientation at the metaphase plate; the possible number of alignments therefore equals 2^n, where n is the number of chromosomes per set. Humans have 23 chromosome pairs, which results in over eight million (2^{23}) possible genetically-distinct gametes. This number does not include the variability that was previously created in the sister chromatids by crossover. Given these two mechanisms, it is highly unlikely that any two haploid cells resulting from meiosis will have the same genetic composition.





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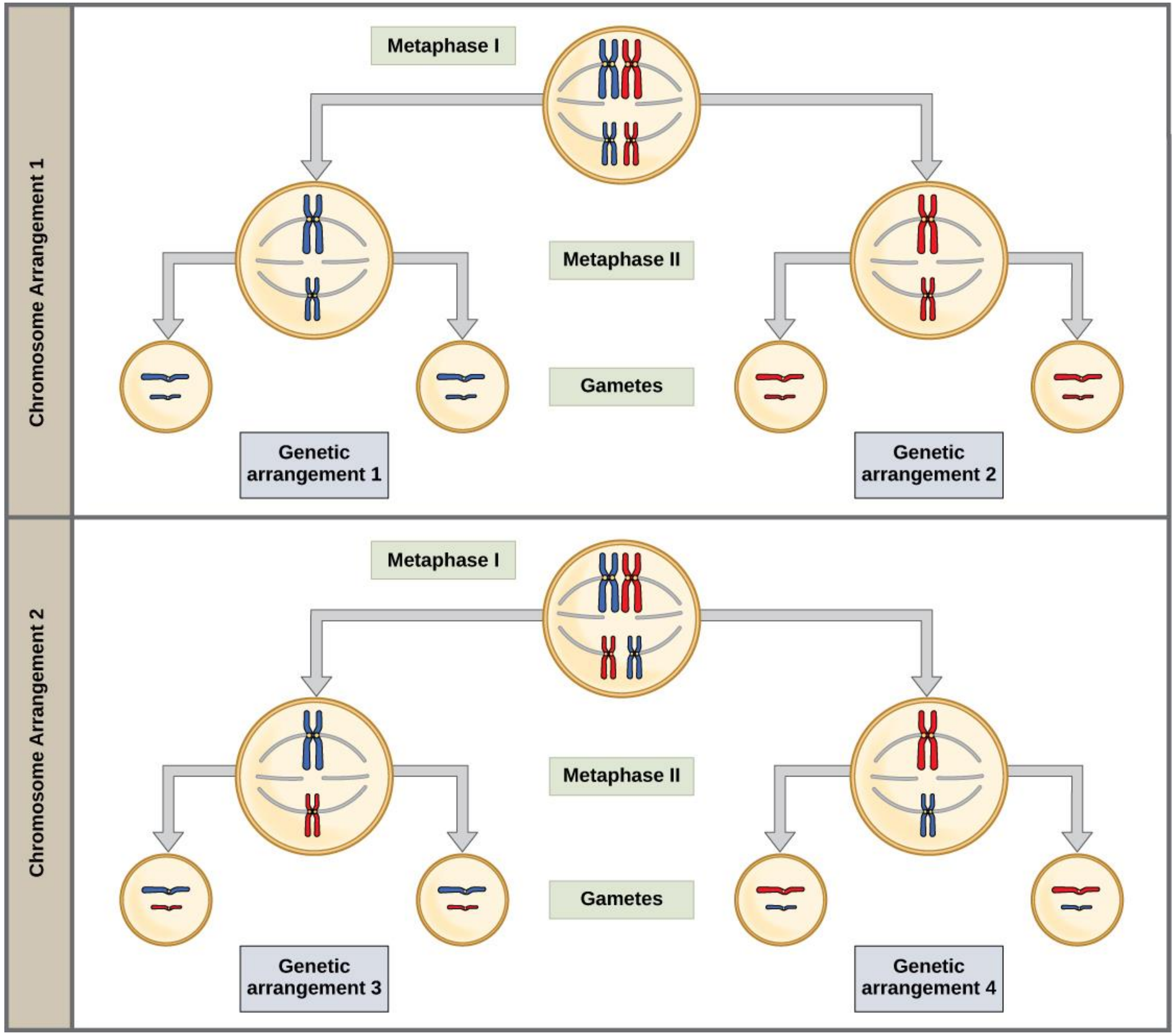
Meiosis I: Metaphase I



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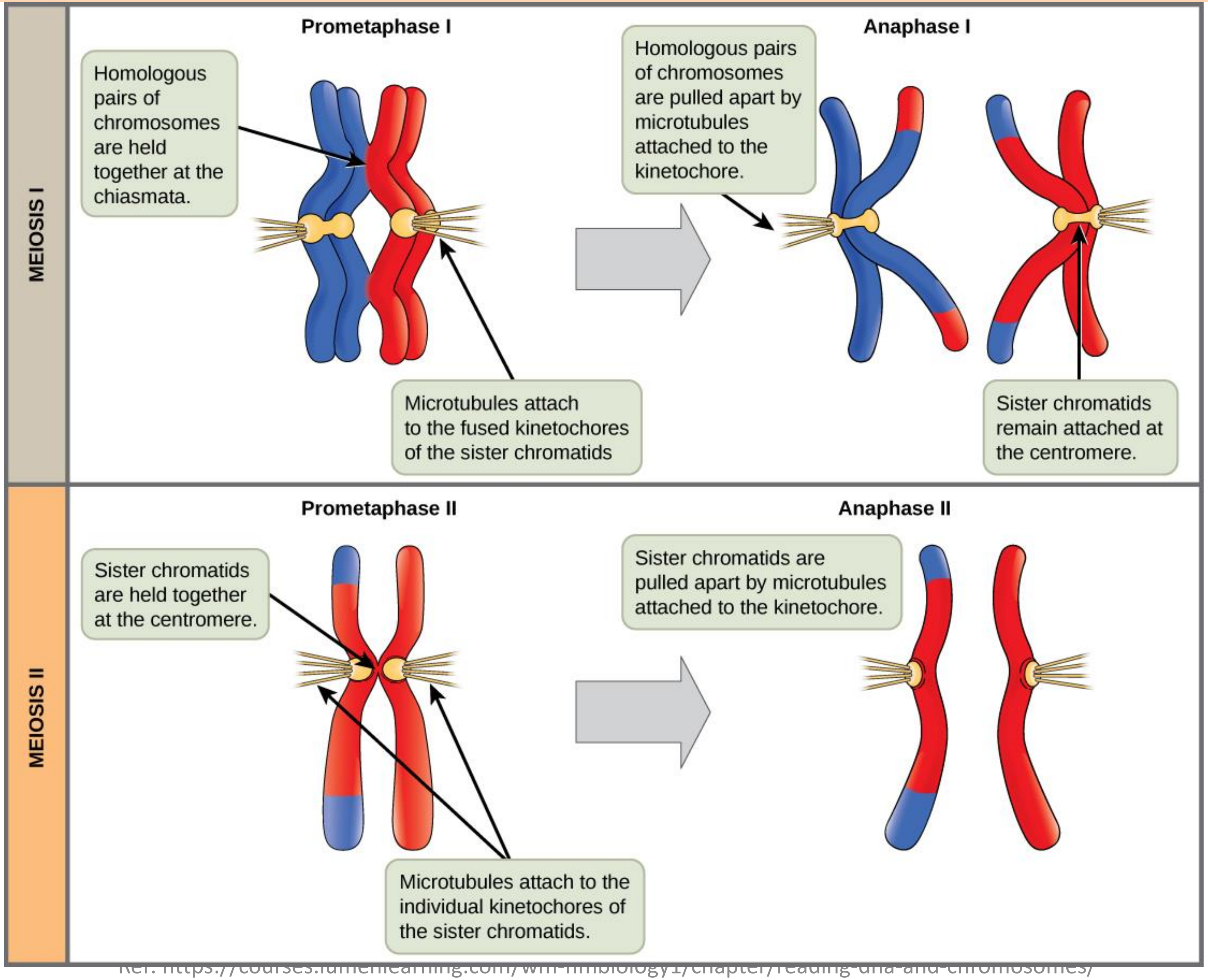
Meiosis I: Anaphase I



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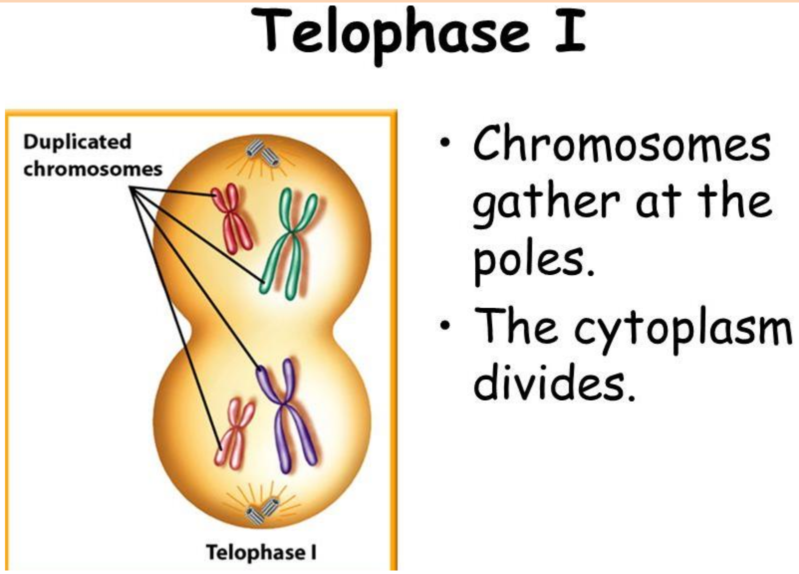
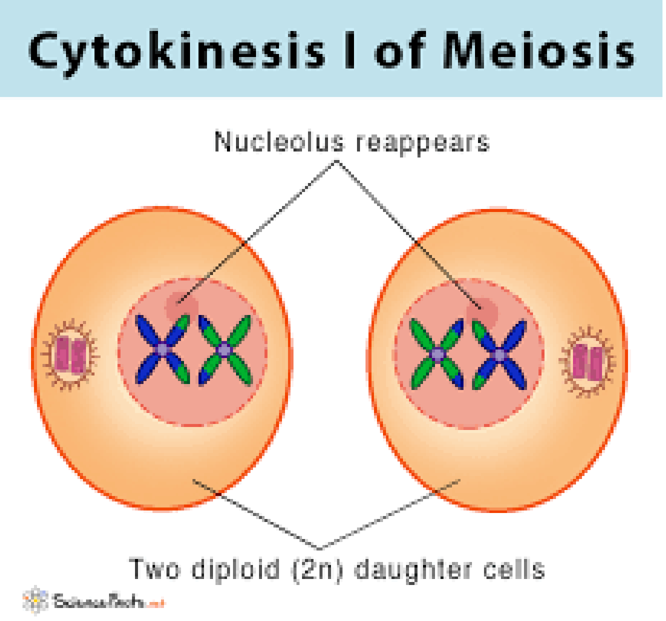
Meiosis I: Telophase I and Cytokinesis

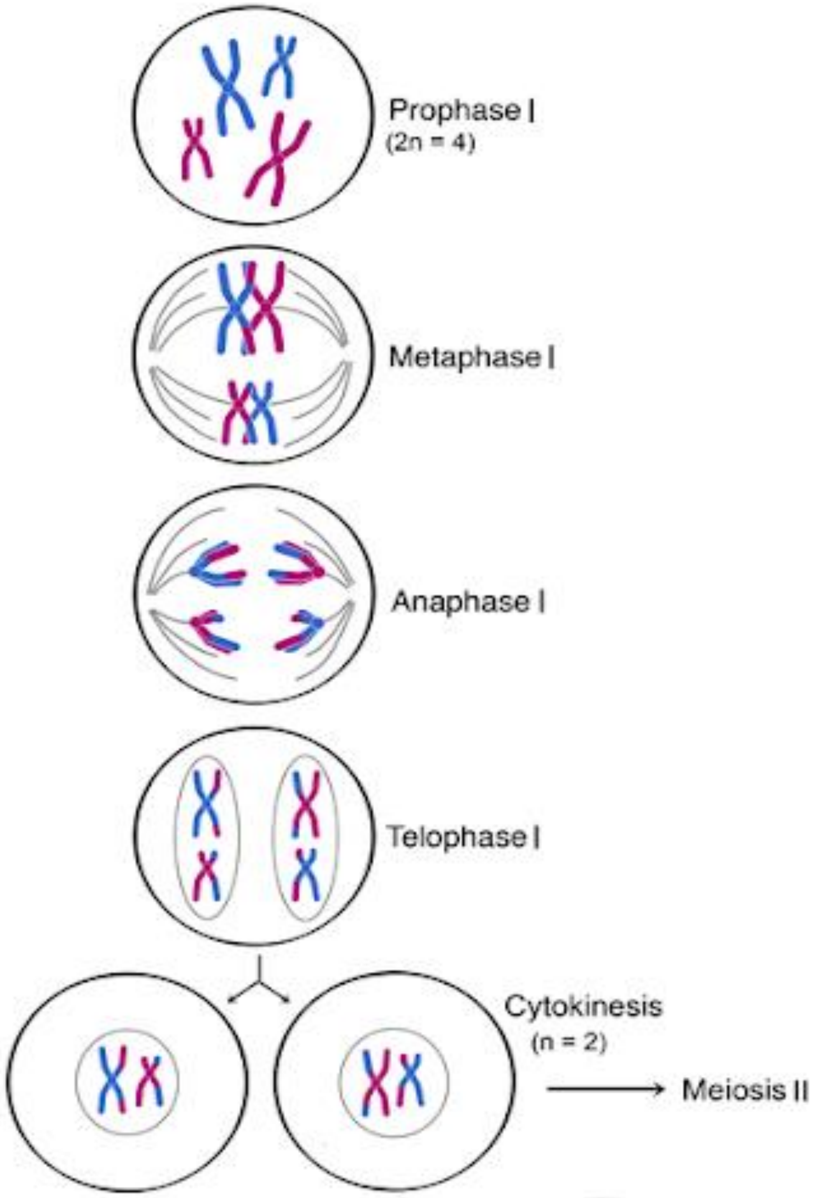
- In telophase, the separated chromosomes arrive at opposite poles. The remainder of the typical telophase events may or may not occur, depending on the species. In some organisms, the chromosomes decondense and nuclear envelopes form around the chromatids in telophase I. In other organisms, cytokinesis—the physical separation of the cytoplasmic components into two daughter cells—occurs without reformation of the nuclei. In nearly all species of animals and some fungi, cytokinesis separates the cell contents via a cleavage furrow (constriction of the actin ring that leads to cytoplasmic division). In plants, a cell plate is formed during cell cytokinesis by Golgi vesicles fusing at the metaphase plate. This cell plate will ultimately lead to the formation of cell walls that separate the two daughter cells.





Two haploid cells are the end result of the first meiotic division

- The cells are haploid because at each pole, there is just one of each pair of the homologous chromosomes. Therefore, only one full set of the chromosomes is present. This is why the cells are considered haploid—there is only one chromosome set, even though each homolog still consists of two sister chromatids. Recall that sister chromatids are merely duplicates of one of the two homologous chromosomes (except for changes that occurred during crossing over). In meiosis II, these two sister chromatids will separate, creating four haploid daughter cells.



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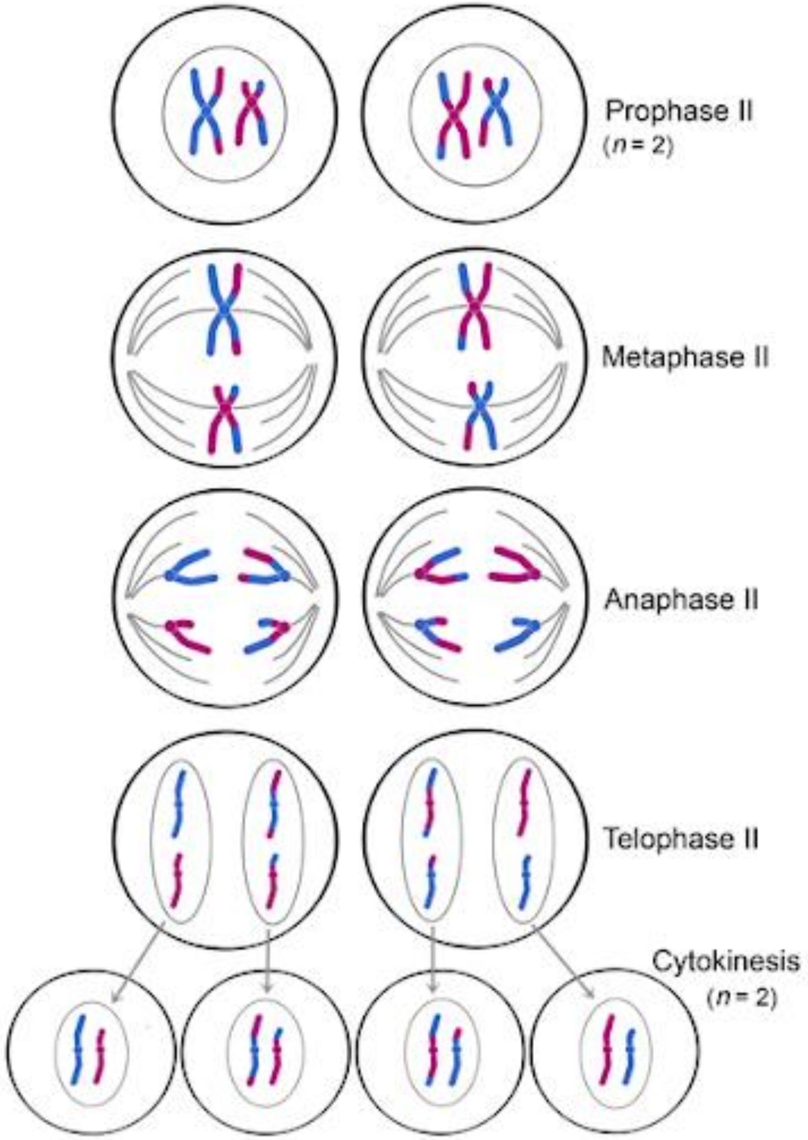
Meiosis II

In some species, cells enter a brief interphase, or **interkinesis**, before entering meiosis II. Interkinesis lacks an S phase, so chromosomes are not duplicated. The two cells produced in meiosis I go through the events of meiosis II in synchrony. During meiosis II, the sister chromatids within the two daughter cells separate, forming four new haploid gametes. The mechanics of meiosis II is similar to mitosis, except that each dividing cell has only one set of homologous chromosomes. Therefore, each cell has half the number of sister chromatids to separate out as a diploid cell undergoing mitosis.



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Meiosis II

Prophase II

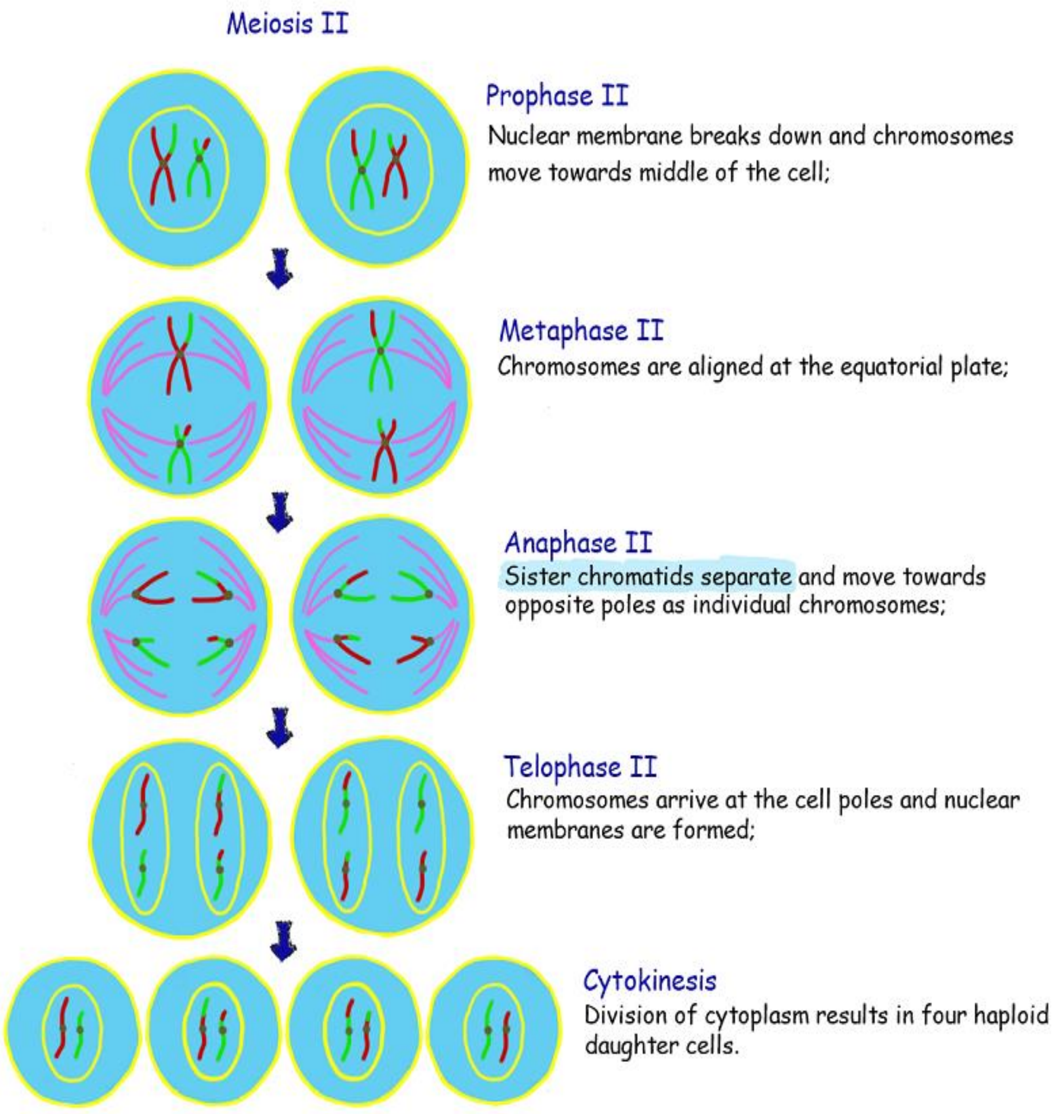
- If the chromosomes decondensed in telophase I, they condense again. If nuclear envelopes were formed, they fragment into vesicles. The centrosomes that were duplicated during interkinesis move away from each other toward opposite poles, and new spindles are formed. The nuclear envelopes are completely broken down, and the spindle is fully formed. Each sister chromatid forms an individual kinetochore that attaches to microtubules from opposite poles.

Metaphase II

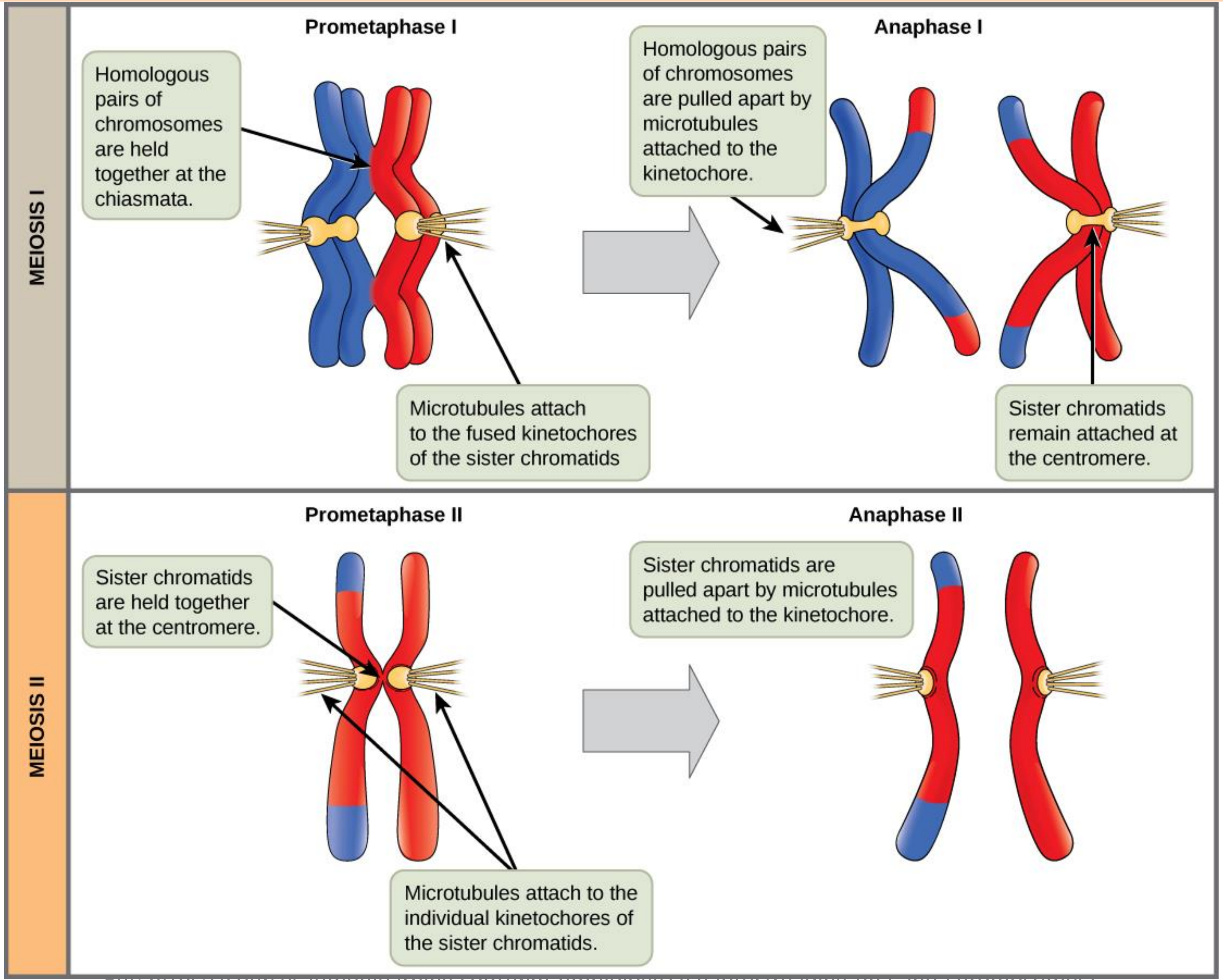
- The sister chromatids are maximally condensed and aligned at the equator of the cell.

Anaphase II

- The sister chromatids are pulled apart by the kinetochore microtubules and move toward opposite poles. Non-kinetochore microtubules elongate the cell.



Meiosis II



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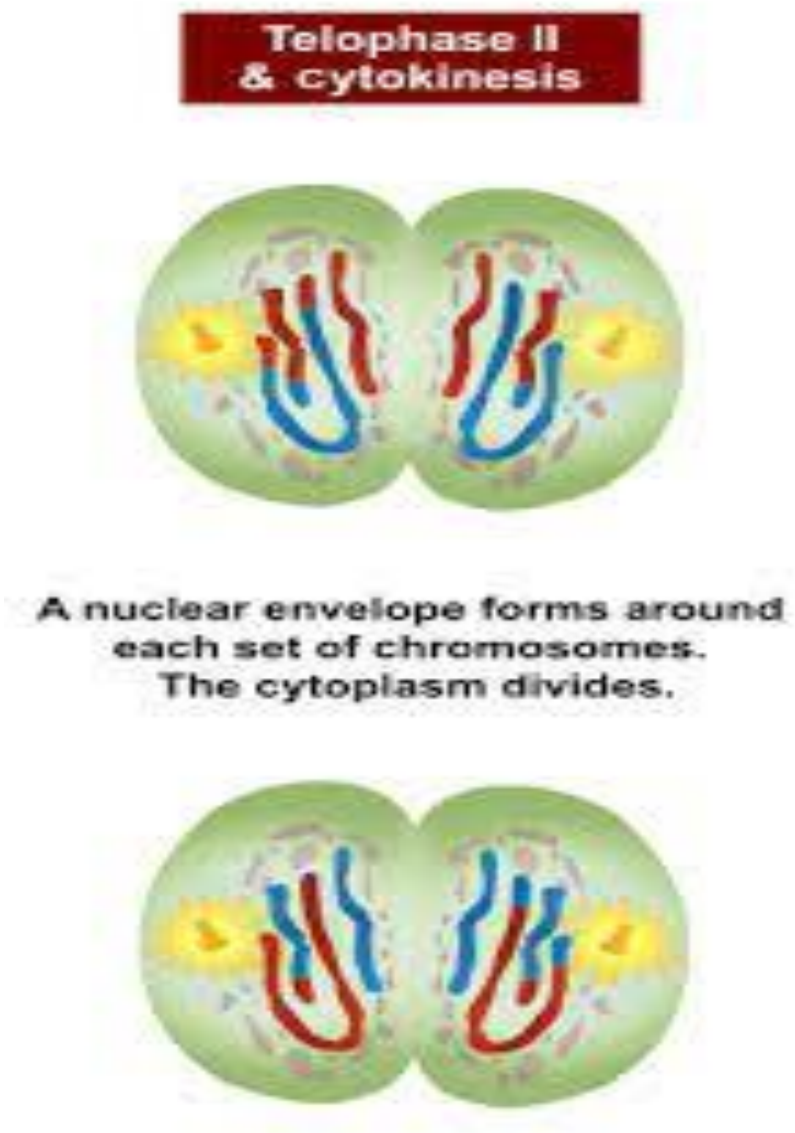
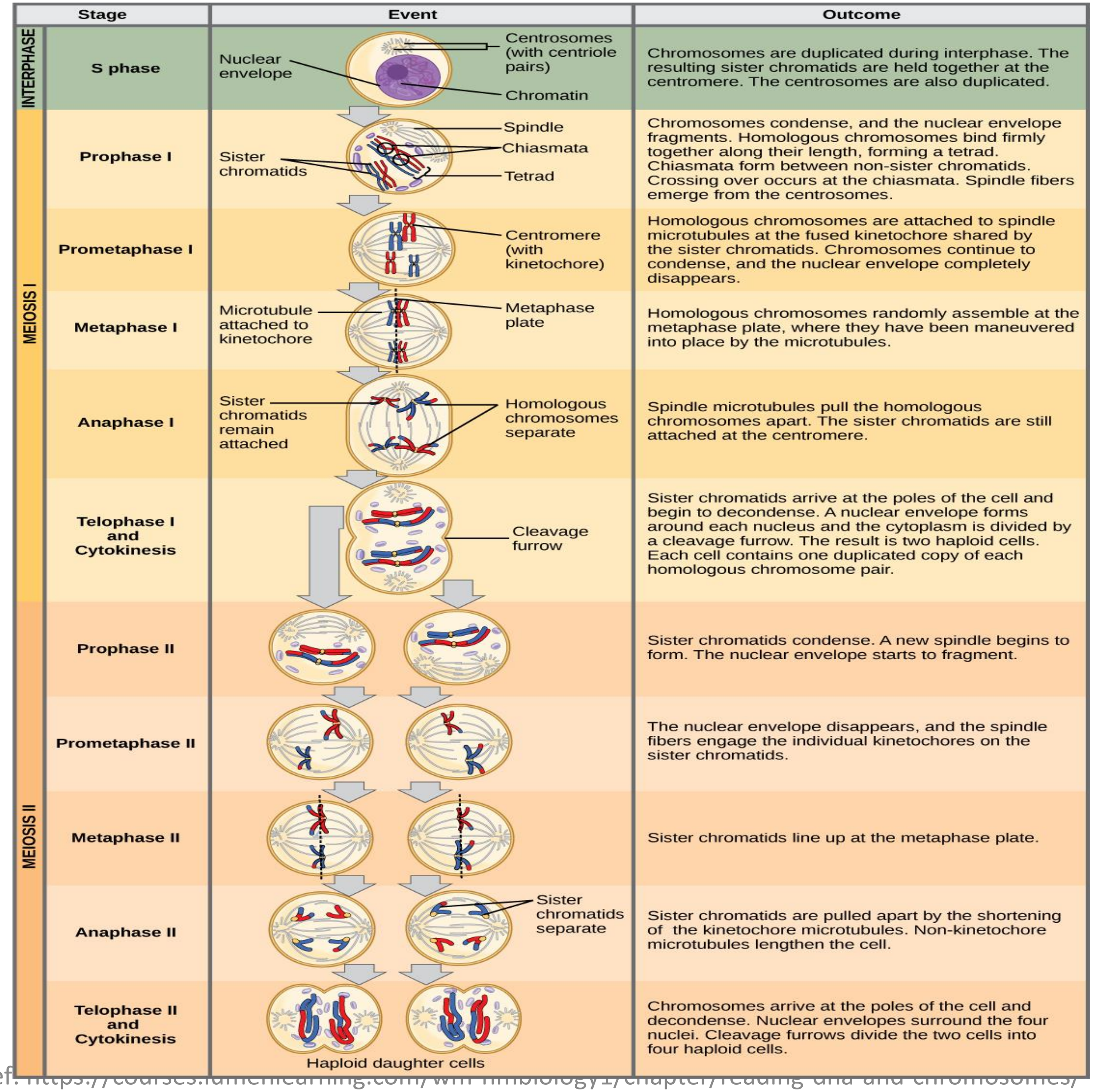
Meiosis II

Telophase II and Cytokinesis

The chromosomes arrive at opposite poles and begin to decondense. Nuclear envelopes form around the chromosomes. Cytokinesis separates the two cells into four unique haploid cells. At this point, the newly formed nuclei are both haploid. The cells produced are genetically unique because of the random assortment of paternal and maternal homologs and because of the recombining of maternal and paternal segments of chromosomes (with their sets of genes) that occurs during crossover. The entire process of meiosis is outlined



Meiosis: Summary



	Stage	Event	Outcome
INTERPHASE	S phase	Nuclear envelope; Centrosomes (with centriole pairs); Chromatin	Chromosomes are duplicated during interphase. The resulting sister chromatids are held together at the centromere. The centrosomes are also duplicated.
MEIOSIS I	Prophase I	Sister chromatids; Spindle; Chiasmata; Tetrad	Chromosomes condense, and the nuclear envelope fragments. Homologous chromosomes bind firmly together along their length, forming a tetrad. Chiasmata form between non-sister chromatids. Crossing over occurs at the chiasmata. Spindle fibers emerge from the centrosomes.
MEIOSIS I	Prometaphase I	Centromere (with kinetochore)	Homologous chromosomes are attached to spindle microtubules at the fused kinetochore shared by the sister chromatids. Chromosomes continue to condense, and the nuclear envelope completely disappears.
MEIOSIS I	Metaphase I	Microtubule attached to kinetochore; Metaphase plate	Homologous chromosomes randomly assemble at the metaphase plate, where they have been maneuvered into place by the microtubules.
MEIOSIS I	Anaphase I	Sister chromatids remain attached; Homologous chromosomes separate	Spindle microtubules pull the homologous chromosomes apart. The sister chromatids are still attached at the centromere.
MEIOSIS I	Telophase I and Cytokinesis	Cleavage furrow	Sister chromatids arrive at the poles of the cell and begin to decondense. A nuclear envelope forms around each nucleus and the cytoplasm is divided by a cleavage furrow. The result is two haploid cells. Each cell contains one duplicated copy of each homologous chromosome pair.
MEIOSIS II	Prophase II		Sister chromatids condense. A new spindle begins to form. The nuclear envelope starts to fragment.
MEIOSIS II	Prometaphase II		The nuclear envelope disappears, and the spindle fibers engage the individual kinetochores on the sister chromatids.
MEIOSIS II	Metaphase II		Sister chromatids line up at the metaphase plate.
MEIOSIS II	Anaphase II	Sister chromatids separate	Sister chromatids are pulled apart by the shortening of the kinetochore microtubules. Non-kinetochore microtubules lengthen the cell.
MEIOSIS II	Telophase II and Cytokinesis	Haploid daughter cells	Chromosomes arrive at the poles of the cell and decondense. Nuclear envelopes surround the four nuclei. Cleavage furrows divide the two cells into four haploid cells.

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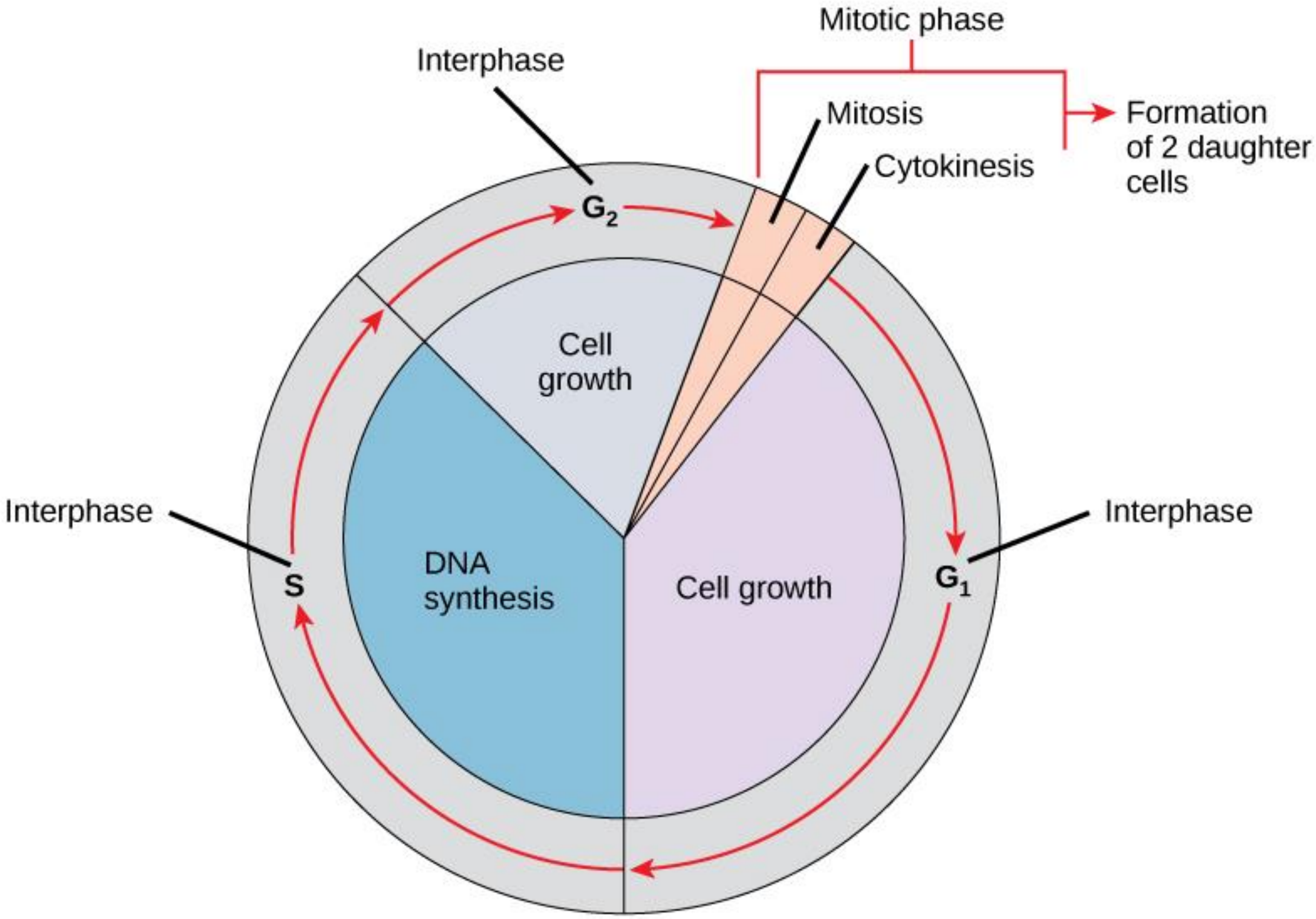
The Complete Cell Cycle



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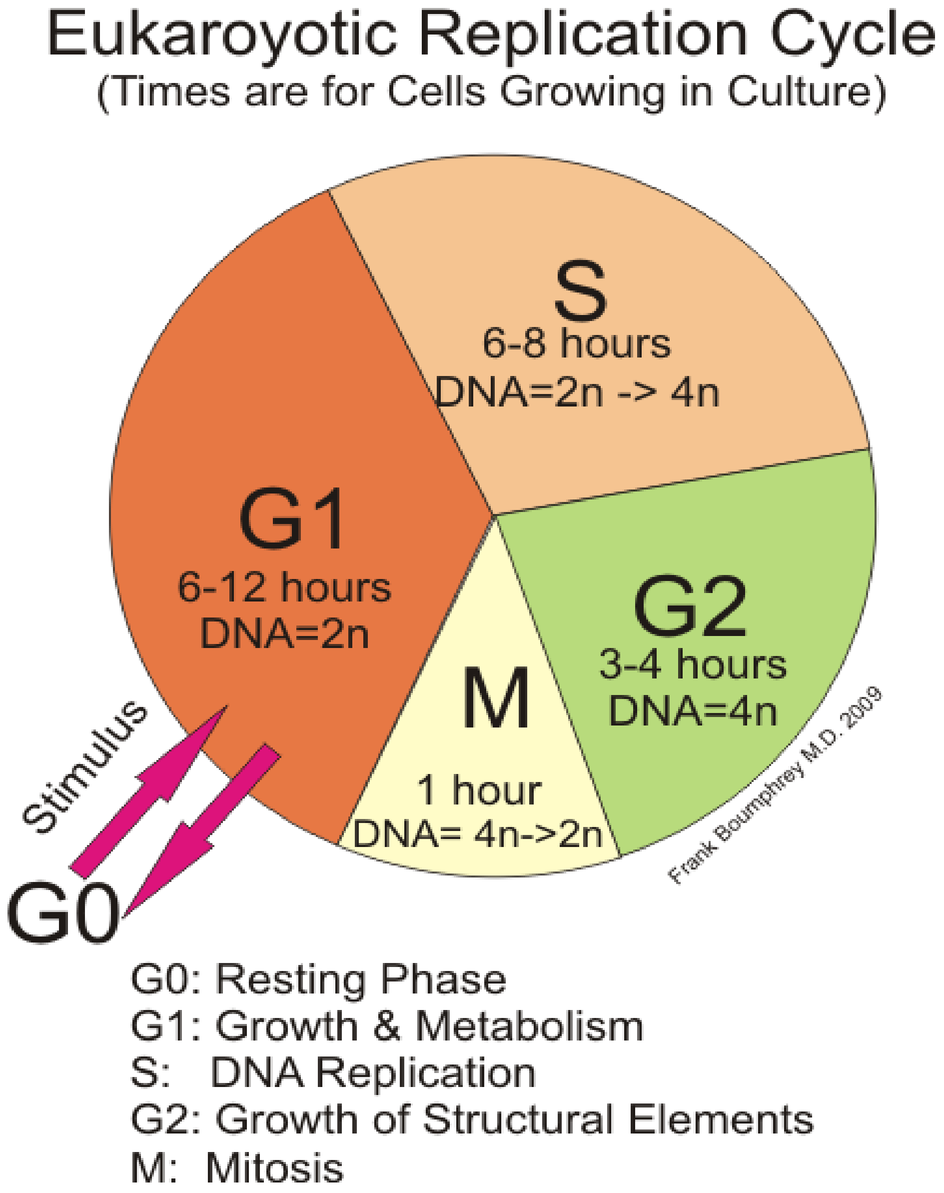
Control of the Cell Cycle

The length of the cell cycle is highly variable, even within the cells of a single organism. In humans, the frequency of cell turnover ranges from a few hours in early embryonic development, to an average of two to five days for epithelial cells, and to an entire human lifetime spent in G_0 by specialized cells, such as cortical neurons or cardiac muscle cells. There is also variation in the time that a cell spends in each phase of the cell cycle.

When fast-dividing mammalian cells are grown in culture (outside the body under optimal growing conditions), the length of the cycle is about 24 hours. In rapidly dividing human cells with a 24-hour cell cycle, the G_1 phase lasts approximately nine hours, the S phase lasts 10 hours, the G_2 phase lasts about four and one-half hours, and the M phase lasts approximately one-half hour. In early embryos of fruit flies, the cell cycle is completed in about eight minutes. The timing of events in the cell cycle is controlled by mechanisms that are both internal and external to the cell.



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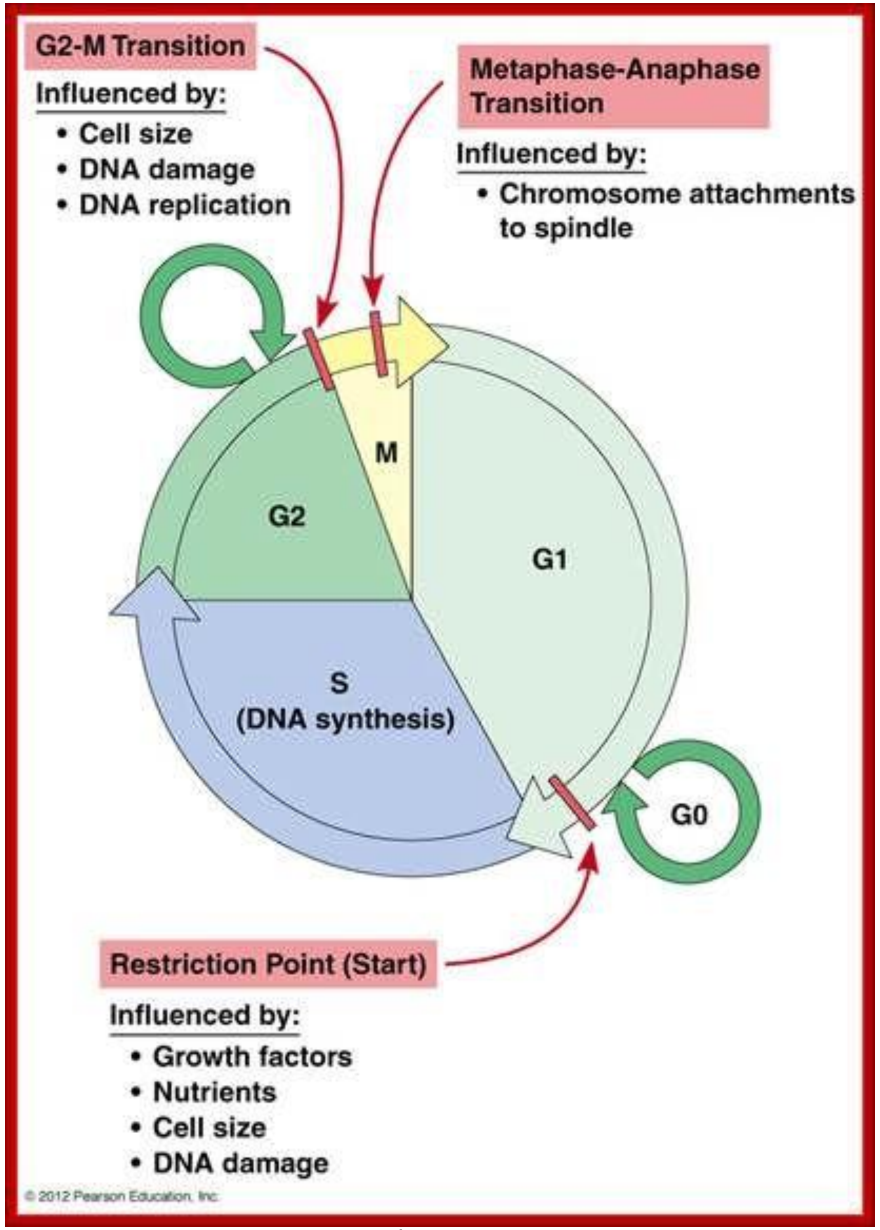
Control of the Cell Cycle

Regulation of the Cell Cycle by External Events:

- Both **the initiation and inhibition** of cell division are triggered by events external to the cell when it is about to begin the replication process.
- the death of a nearby cell or as sweeping as the release of growth-promoting hormones, such as **human growth hormone** (HGH).
- A lack of HGH can inhibit cell division, resulting in dwarfism, whereas too much HGH can result in gigantism.
- **Crowding of** cells can also inhibit cell division. Another factor that can initiate cell division is the size of the cell; as a cell grows, it becomes inefficient due to its decreasing surface-to-volume ratio. The solution to this problem is to divide.
- Whatever the source of the message, the cell receives the signal, and a series of events within the cell allows it to proceed into mitosis. Moving forward from this initiation point, every parameter required during each cell cycle phase must be met or the cycle cannot progress.



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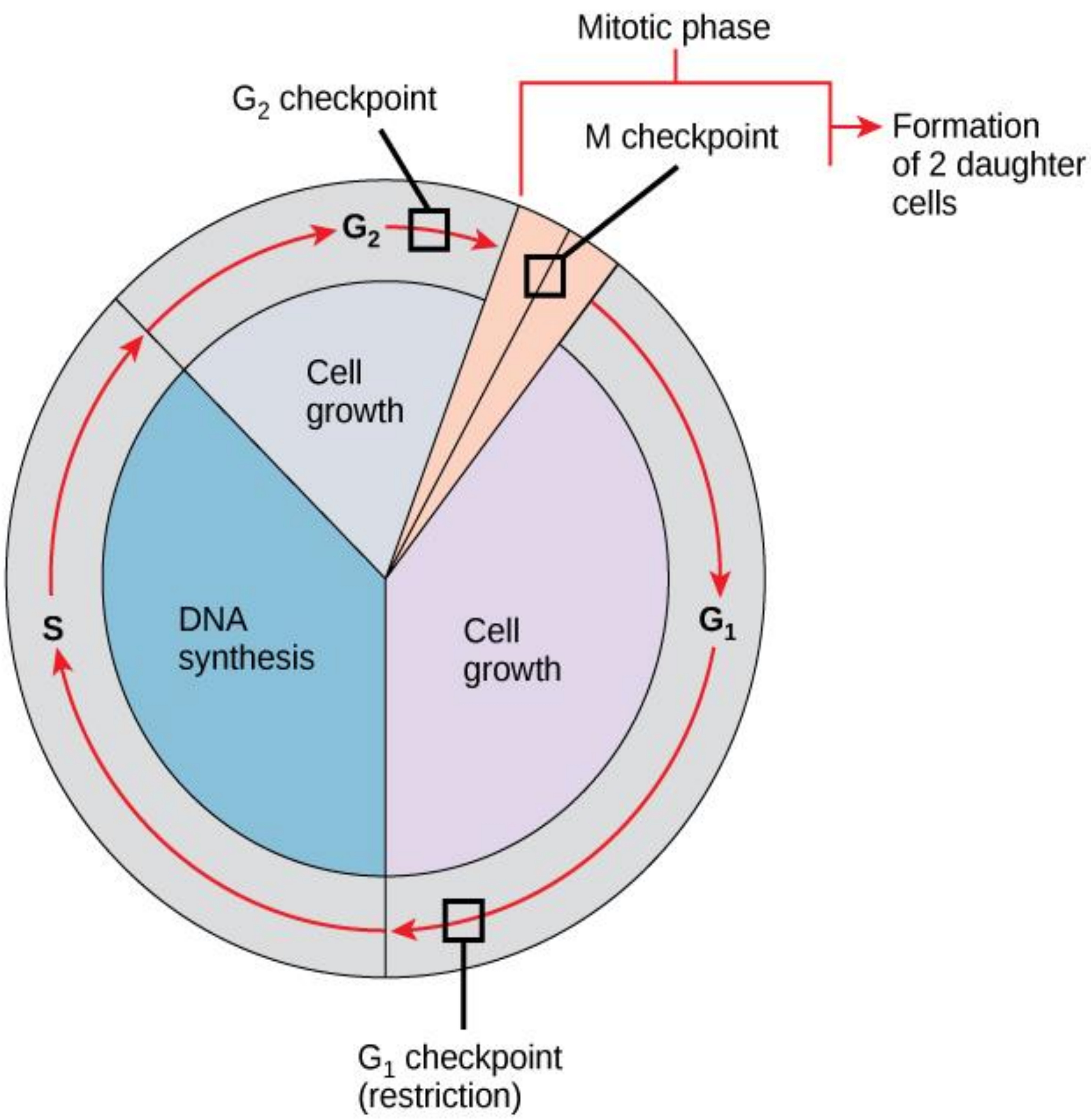
Control of the Cell Cycl2e

Regulation at Internal Checkpoints

- It is essential that the daughter cells produced be exact duplicates of the parent cell.
- Mistakes in the duplication or distribution of the chromosomes lead to mutations : an abnormal cell.
- To prevent a compromised cell from continuing to divide, there are internal control mechanisms that operate at three main cell cycle checkpoints.
- A checkpoint is one of several points in the eukaryotic cell cycle at which the progression of a cell to the next stage in the cycle can be halted until conditions are favorable.
- These checkpoints occur near the end of G_1, at the G_2/M transition, and during metaphase.



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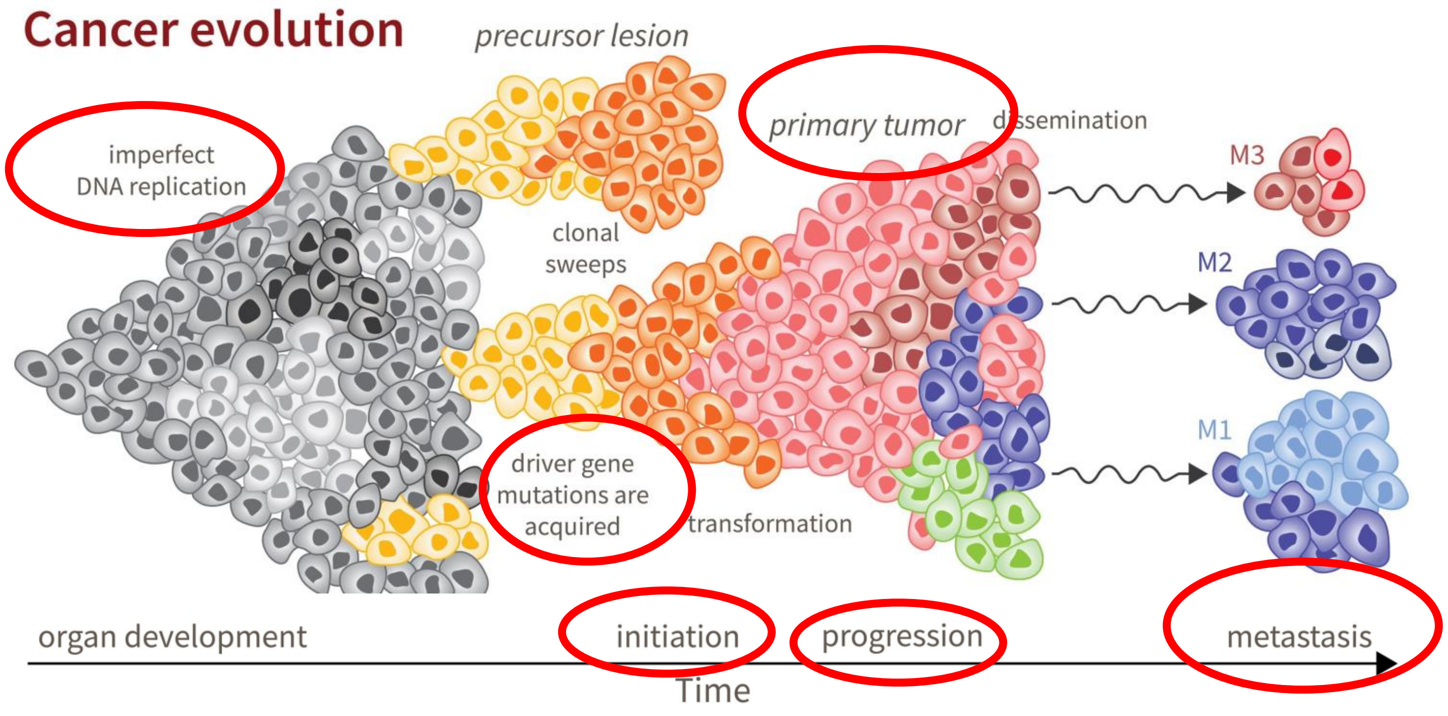
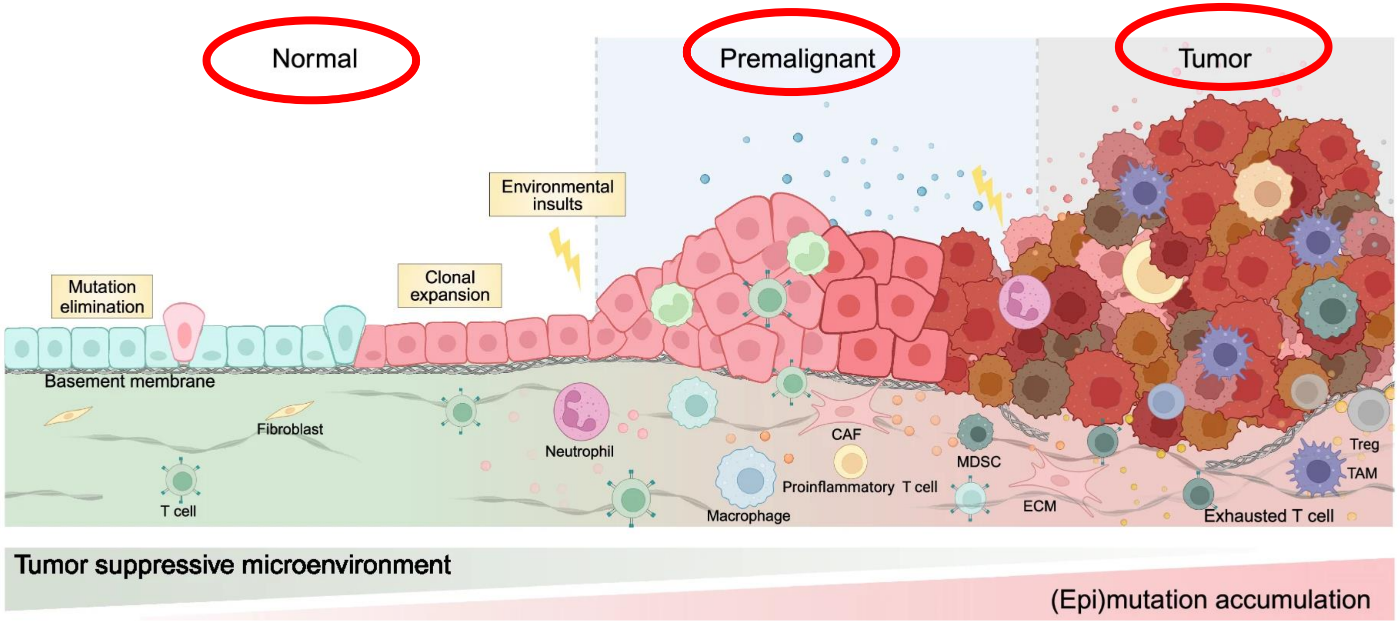
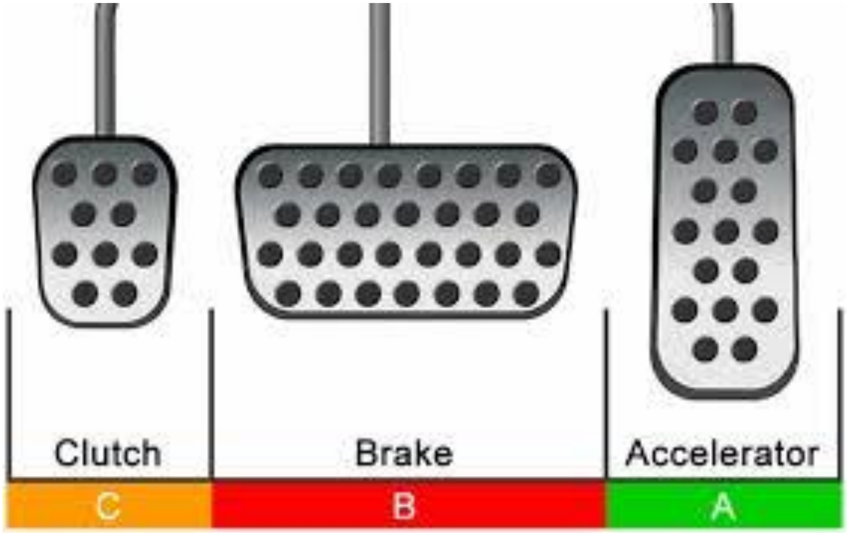
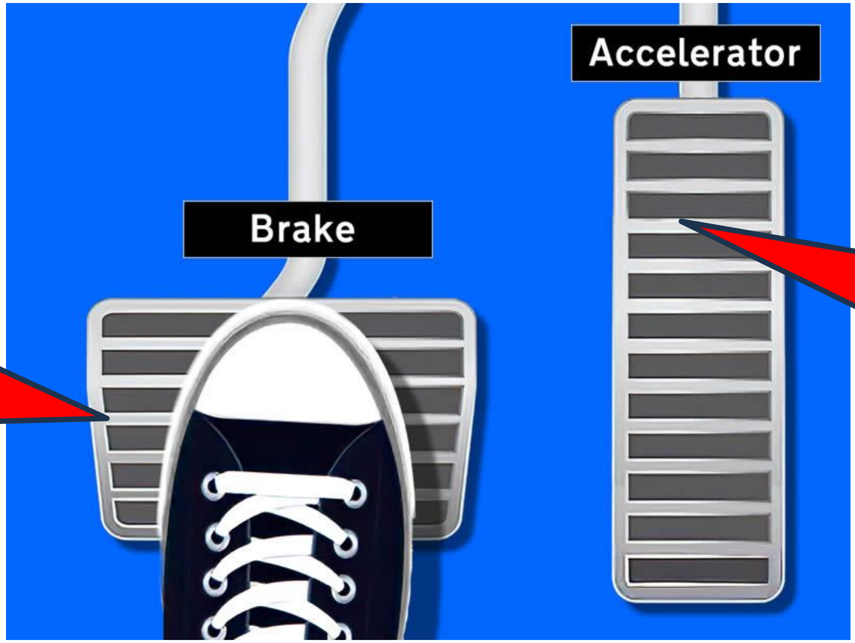
Cancer and the Cell Cycle





Proto-oncogenes, Oncogenes, Tumor Suppressor Genes

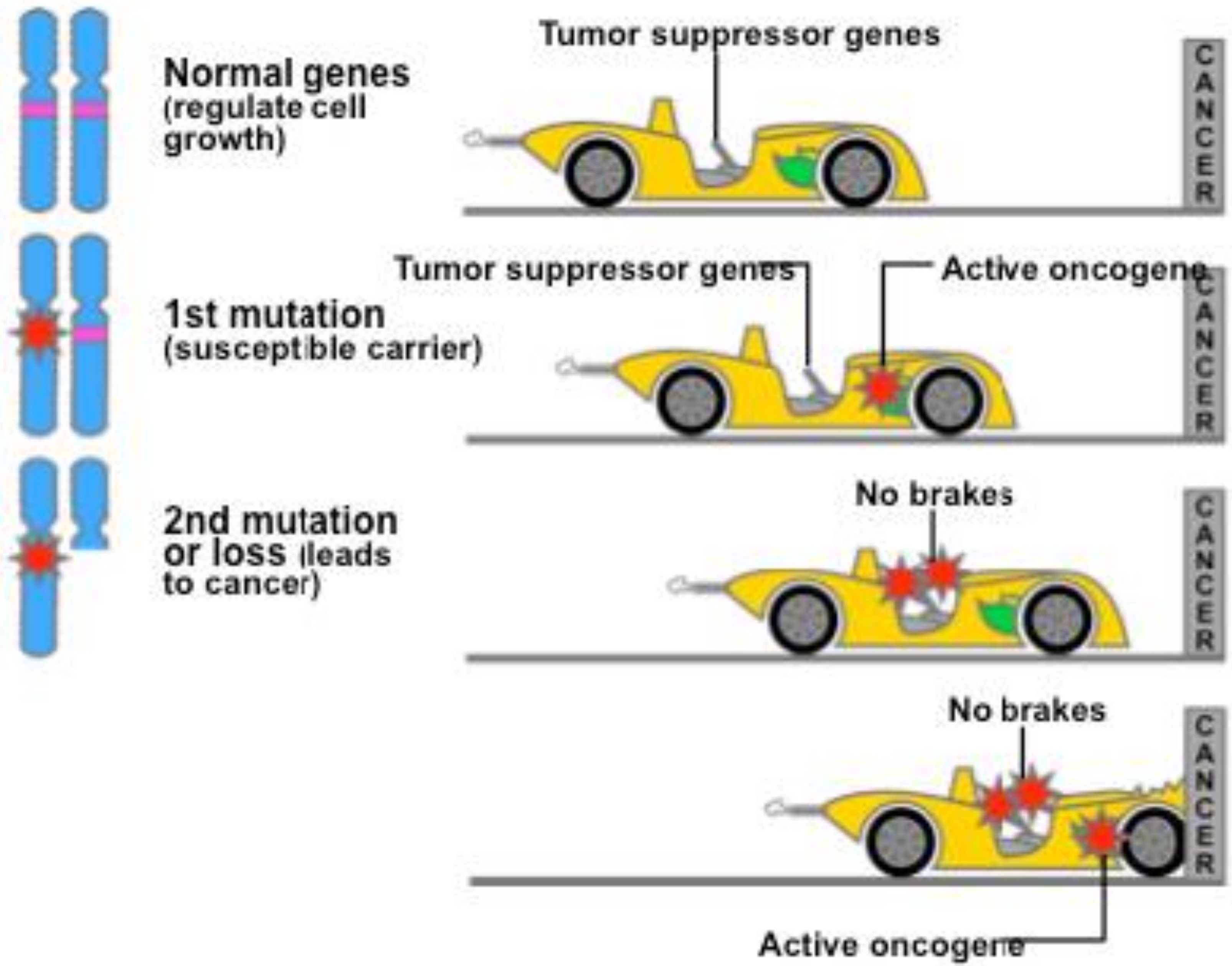




TP53, APC, BRCA1, BRCA2

KRAS
CMYC
EGFR

Proto-oncogenes, Oncogenes, Tumor Suppressor Genes



Proto-oncogenes

Genes that normally help cells grow and divide to make new cells, or to help cells stay alive. When a proto-oncogene mutates (changes) or there are too many copies of it, it can become turned on (activated) when it is not supposed to be, at which point it's now called an **oncogene**. When this happens, the cell can start to grow out of control, which might lead to cancer.

A proto-oncogene normally functions in a way much like the gas pedal on a car. It helps the cell grow and divide. An oncogene is like a gas pedal that is stuck down, which causes the cell to divide out of control.

Common oncogenes are:

KRAS,

C-Myc,

EGFR,

PIK3CA

APC

HER2…………..

Cancer Type	Common Oncogenic or Tumor Suppressor Gene Origin
Leukemia, breast, colon, gastric and lung Cancer	c-MYC amplification
Renal cell cancer	Von Hippel-Lindaugene (VHL) dys-function
Chronic myelogenous leukemia	Bcr-abl proto-oncogene translocation
Childhood neuroblastoma and small cell lung Cancer	n-MYC amplification
Follicular lymphoma	Bcl-2 amplification, myc mutation
Sporadic thyroid cancer	Ret mutation
Familial melanoma	$P16^{INK4A}$ mutation
Familial breast and ovarian cancer	BRCA1, BRCA2 mutation
Colorectal and gastric cancer	APC gene mutation
Invasive ductal breast cancer	HER-2 amplification

Tumor suppressor genes

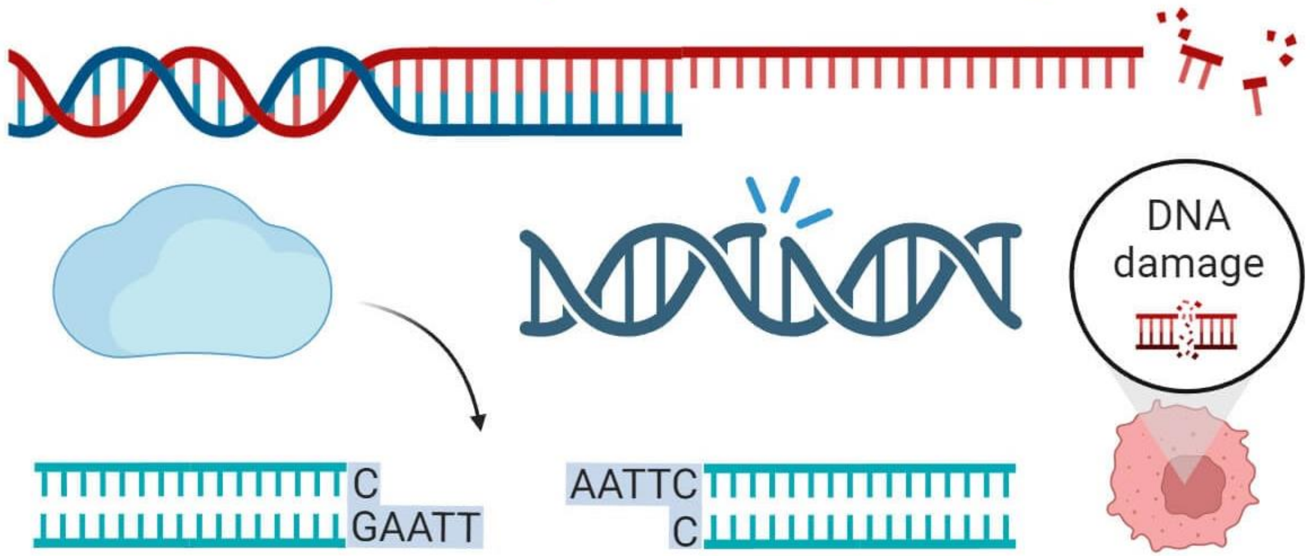
- Genes that **slow down cell division** or tell cells to die at the right time (a process known as *apoptosis* or *programmed cell death*).
- When tumor suppressor genes don't work properly, **cells can grow out of control**, which can lead to cancer.
- A tumor suppressor gene is like the **brake pedal on a car**. It normally helps keep the cell from dividing too quickly, just as a brake keeps a car from going too fast.
- When something goes wrong with a tumor suppressor gene, such as mutation that stops it from working, cell division can get out of control that ultimately leads cancers.

Common tumor suppressor genes are:
TP53,
APC,
BRCA1, BRCA2,
CDH1,
E-cadherin, CDKN2A, INK4a, MEN1,
NF1, PTEN, RB, SMAD4,, TSC1,
TSC2.

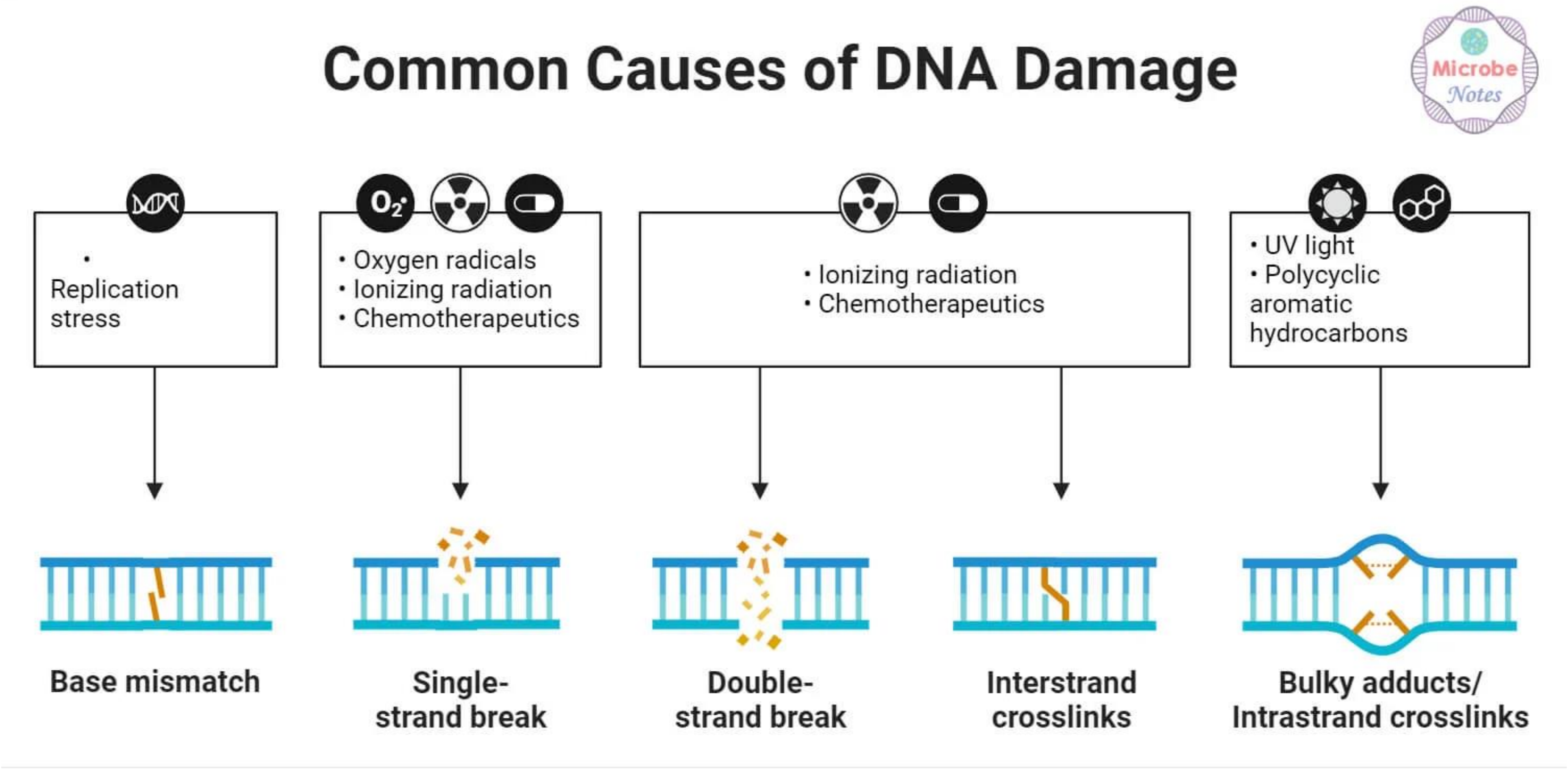
Tumor-Suppressor Gene	Neoplasm(s)
RB1	Retinoblastoma; osteosarcoma; carcinoma of breast, bladder, and lung
p53	Astrocytoma; carcinoma of breast, colon, and lung; osteosarcoma
WT1	Wilms' tumor
DCC	Carcinoma of colon
NF1	Neurofibromatosis type 1
FAP	Carcinoma of colon
MEN-1	Tumors of parathyroid, pancreas, pituitary, and adrenal cortex

DNA Damage (Accidents)
DNA Repair (Car mechanic)

DNA Damage and DNA Repair



- DNA damage repair (DDR) genes are genes that produce proteins that **detect and fix mutations in other genes**. These genes are important for maintaining genomic stability and preventing cancer:
- DDR genes recognize and remove DNA lesions, tolerate DNA damage, and protect against errors during DNA replication or repair
- DNA damage repair (DDR) genes are responsible for maintaining the stability of the genome in cells. When DDR genes are defective, cells are more likely to accumulate DNA mutations, which can lead to cancer
- MESH2, BRCA1, BRCA2, ATM, XRCC5, FEN1, WRN are common DDR genes

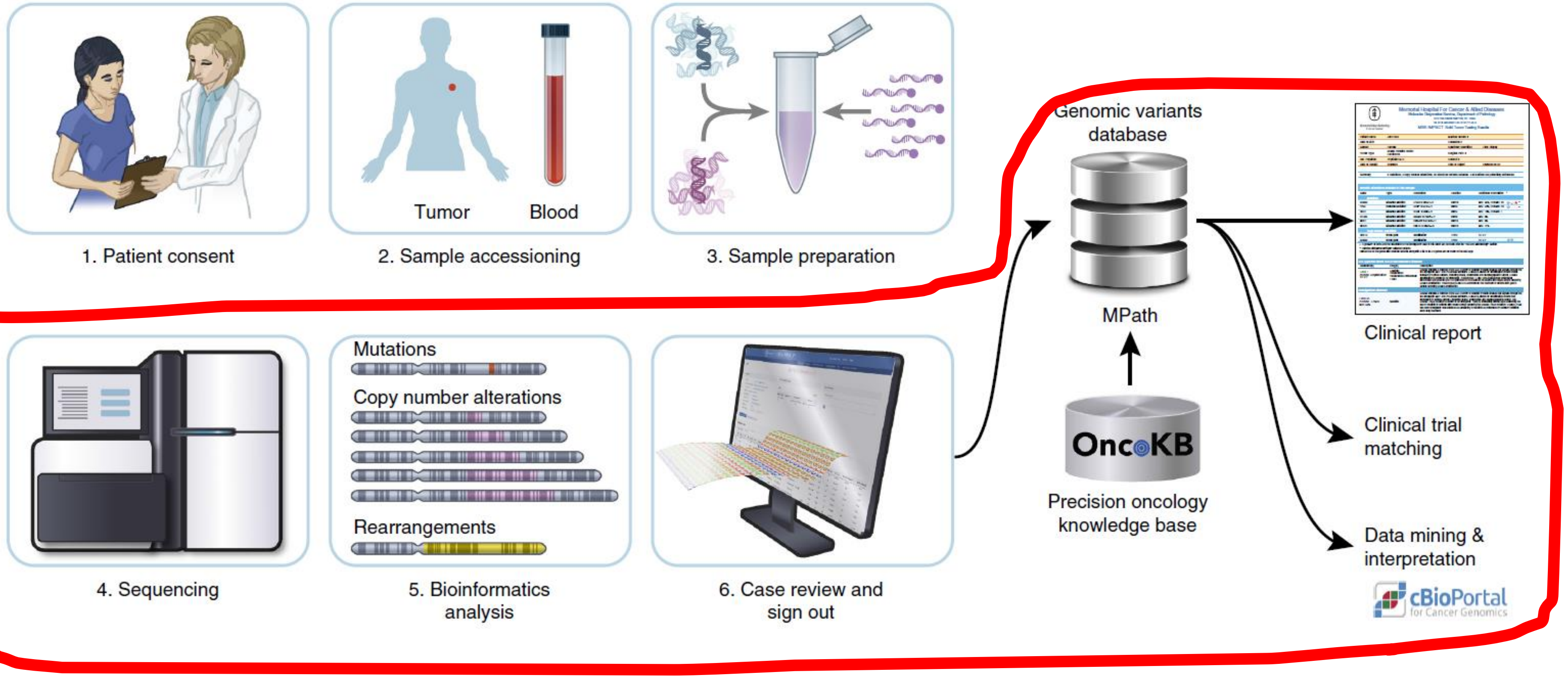


The repair pathway that is used depends on the type of damage that has occurred. For example,

- **Base excision repair (BER),**
- **Nucleotide excision repair (NER),**
- **Non-homologous end joining (NHEJ),** and
- **DNA mismatch repair (MMR)** are all different DNA repair pathways.

Do we (CSE students) really need to know this?

Schematic diagram for the Neoantigen target (mutations) discovery from Hepatocellular carcinoma (HCC) patients:



Mutational landscape of metastatic cancer revealed from prospective clinical sequencing of 10,000 patients

https://www.nature.com/articles/nm.4333

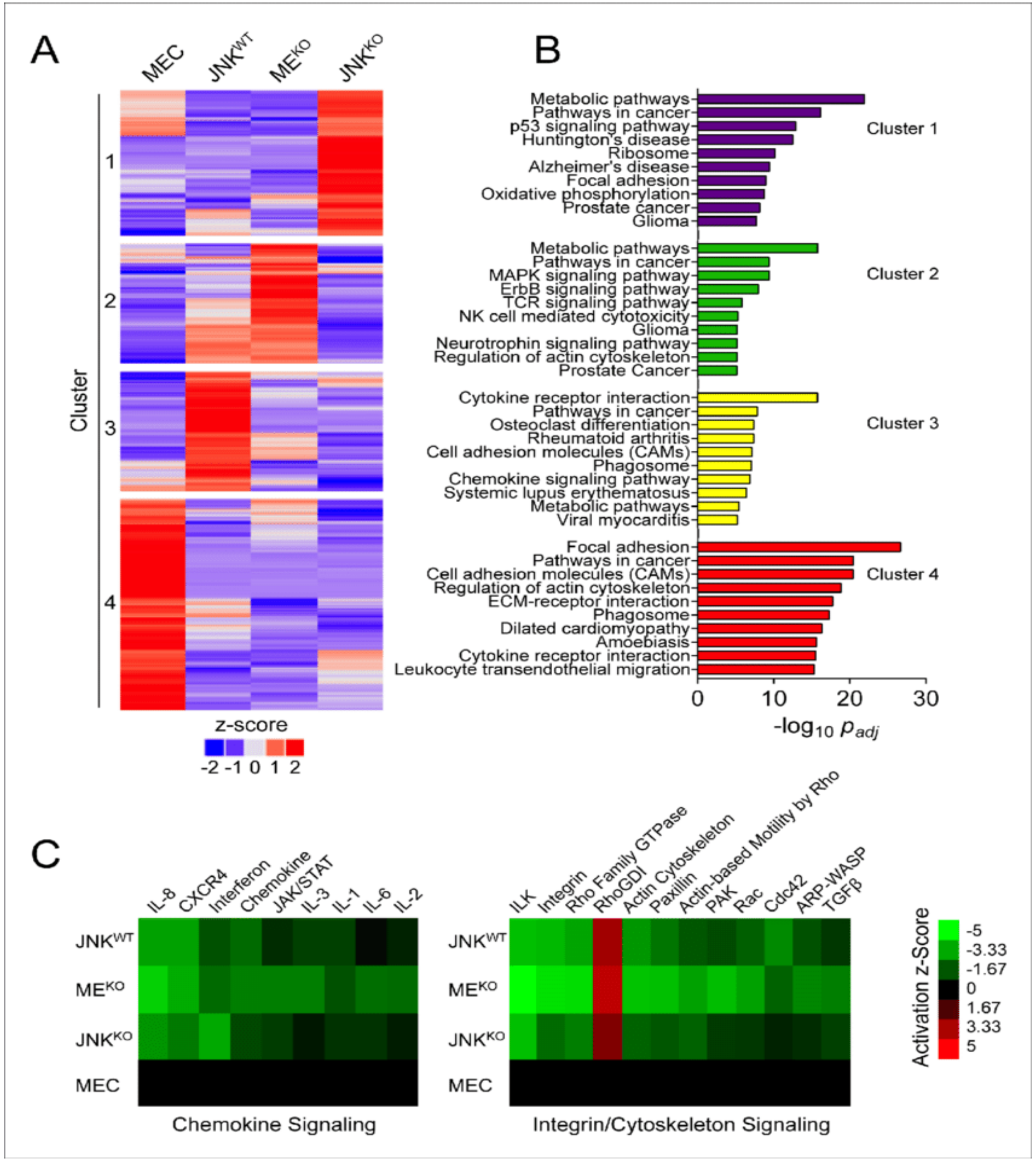
Nature Medicine **volume 23**, pages703–713 (2017)

Why we need to know these stuff !!??

The cJUN NH2-terminal kinase (JNK) signaling pathway promotes genome stability and prevents tumor initiation.

University of Massachusetts Medical School, Worcester, United States.

https://elifesciences.org/articles/36389



https://www.nature.com/articles/s41698-021-00142-x

Do we really need to know this?

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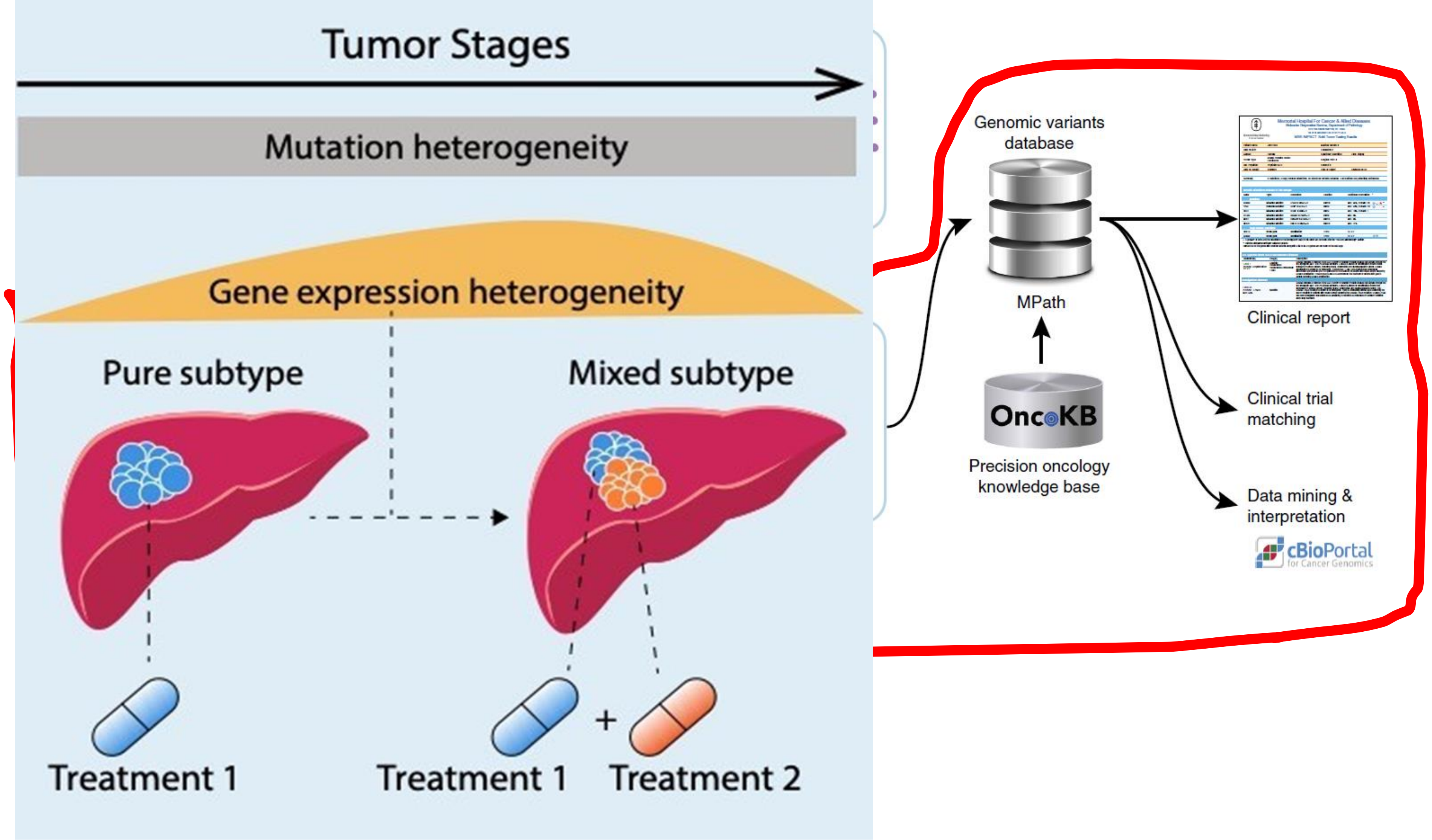
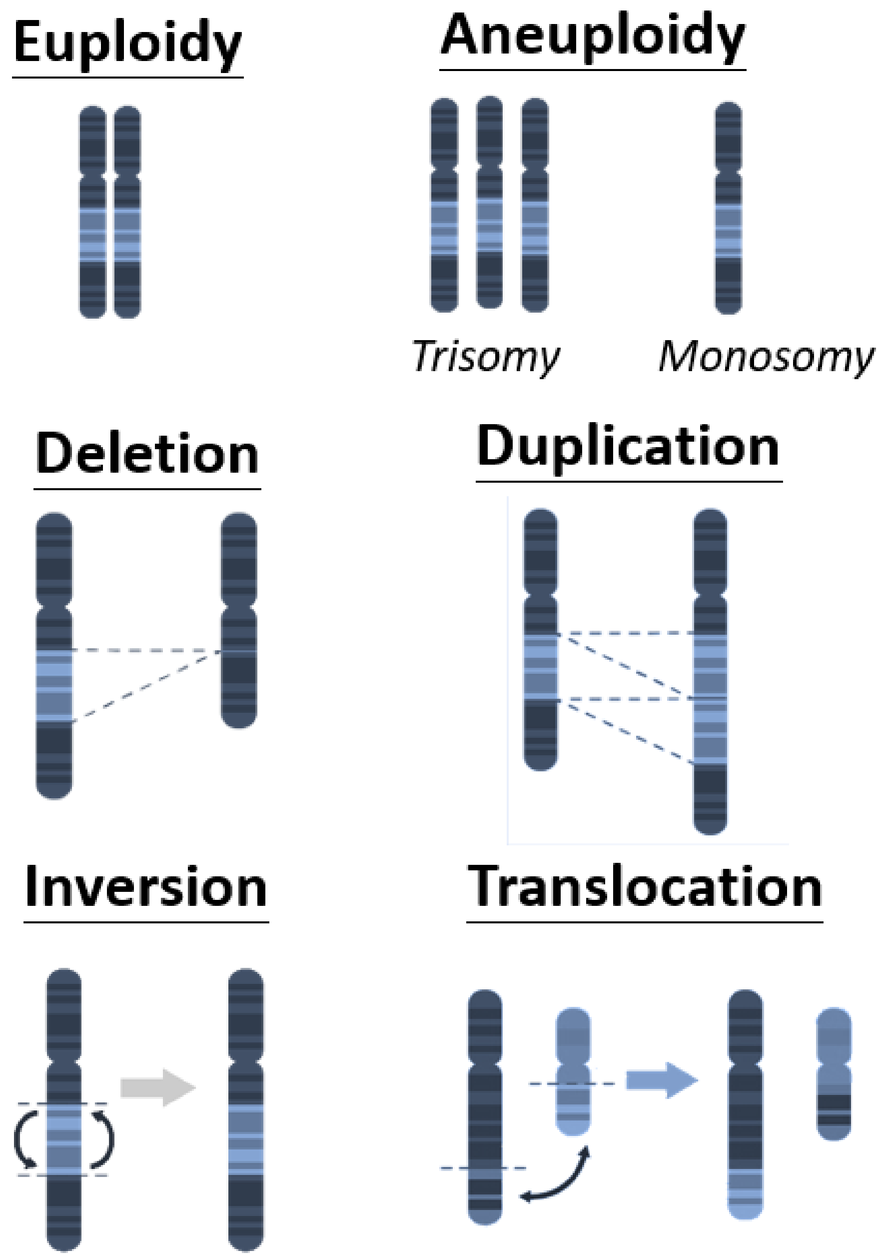


Chromosomal Aberrations

- The four main types of structural chromosomal aberrations are **deletion, duplication, inversion, and translocation**.
- Aneuploidy is the presence of an abnormal number of chromosomes in a cell, for example **a human cell having 45 or 47 chromosomes instead of the usual 46**. It does not include a difference of one or more complete sets of chromosomes.



Common Disorders: Aneuploidy

- An individual with the appropriate number of chromosomes for their species is called **euploid**; in humans, euploidy corresponds to 22 pairs of autosomes and one pair of sex chromosomes. An individual with an error in chromosome number is described as **aneuploid**, a term that includes **monosomy** (loss of one chromosome) or **trisomy** (gain of an extraneous chromosome). Monosomic human zygotes missing any one copy of an autosome invariably fail to develop to birth because they lack essential genes. This underscores the importance of "gene dosage" in humans. Most autosomal trisomies also fail to develop to birth; however, duplications of some of the smaller chromosomes (13, 15, 18, 21, or 22) can result in offspring that survive for several weeks to many years. Trisomic individuals suffer from a different type of genetic imbalance: an excess in gene dose. Individuals with an extra chromosome may synthesize an abundance of the gene products encoded by that chromosome. This extra dose (150 percent) of specific genes can lead to a number of functional challenges and often precludes development. The most common trisomy among viable births is that of chromosome 21, which corresponds to Down Syndrome. Individuals with this inherited disorder are characterized by short stature and stunted digits, facial distinctions that include a broad skull and large tongue, and significant developmental delays. The incidence of Down syndrome is correlated with maternal age; older women are more likely to become pregnant with fetuses carrying the trisomy 21 genotype.

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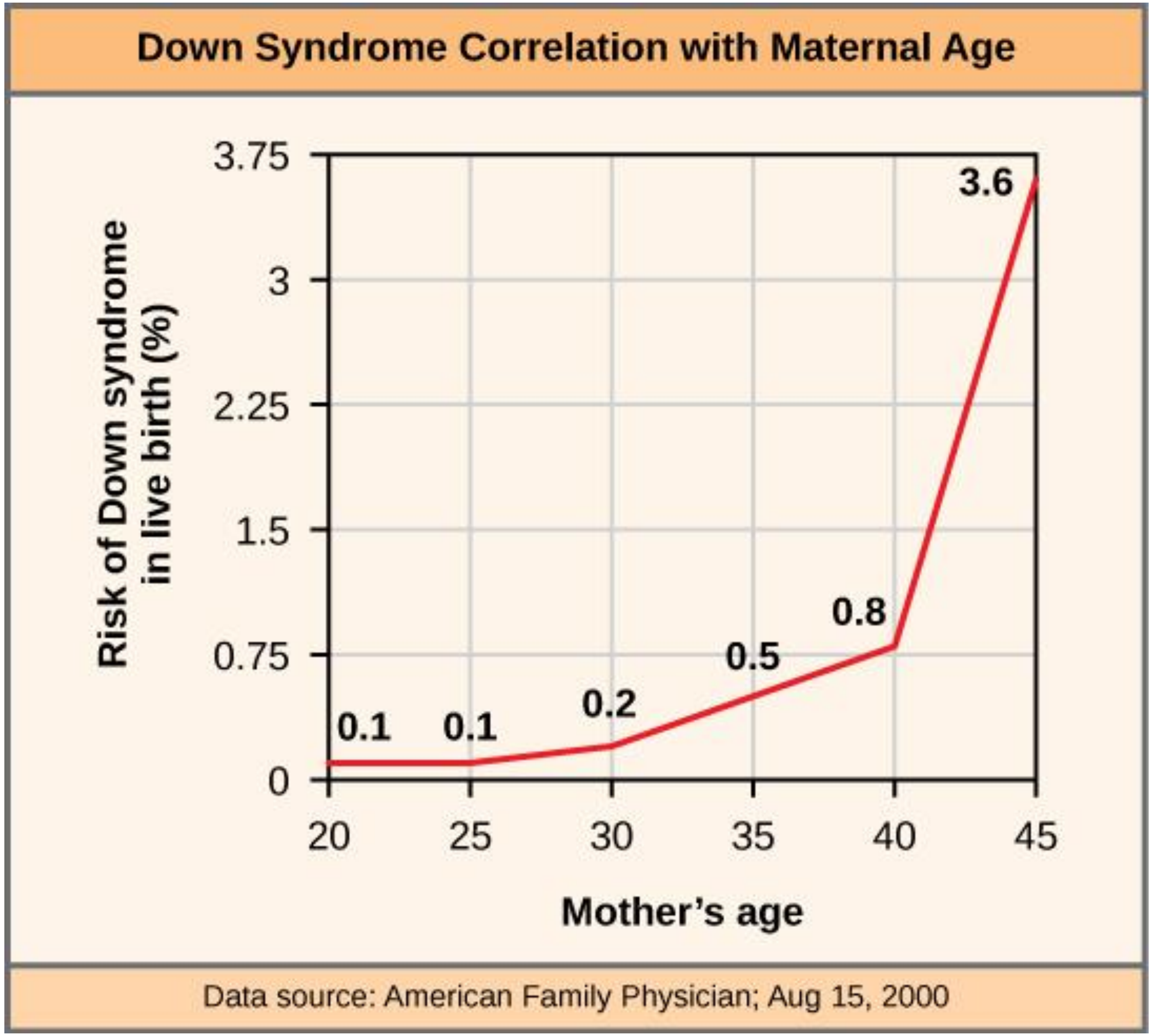
Common Disorders: Aneuploidy



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Common Disorders: Polyploidy

An individual with more than the correct number of chromosome sets (two for diploid species) is called **polyploid**. For instance, fertilization of an abnormal diploid egg with a normal haploid sperm would yield a triploid zygote. Polyploid animals are extremely rare, with only a few examples among the flatworms, crustaceans, amphibians, fish, and lizards. Polyploid animals are sterile because meiosis cannot proceed normally and instead produces mostly aneuploid daughter cells that cannot yield viable zygotes.

Rarely, polyploid animals can reproduce asexually by haplodiploidy, in which an unfertilized egg divides mitotically to produce offspring. In contrast, polyploidy is very common in the plant kingdom, and polyploid plants tend to be larger and more robust than euploids of their species.

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Duplications and Deletions

- In addition to the loss or gain of an entire chromosome, a chromosomal segment may be duplicated or lost. Duplications and deletions often produce offspring that survive but exhibit physical and mental abnormalities. Duplicated chromosomal segments may fuse to existing chromosomes or may be free in the nucleus. Cri-du-chat (from the French for "cry of the cat") is a syndrome associated with nervous system abnormalities and identifiable physical features that result from a deletion of most of 5p (the small arm of chromosome 5) (Figure above). Infants with this genotype emit a characteristic high-pitched cry on which the disorder's name is based.

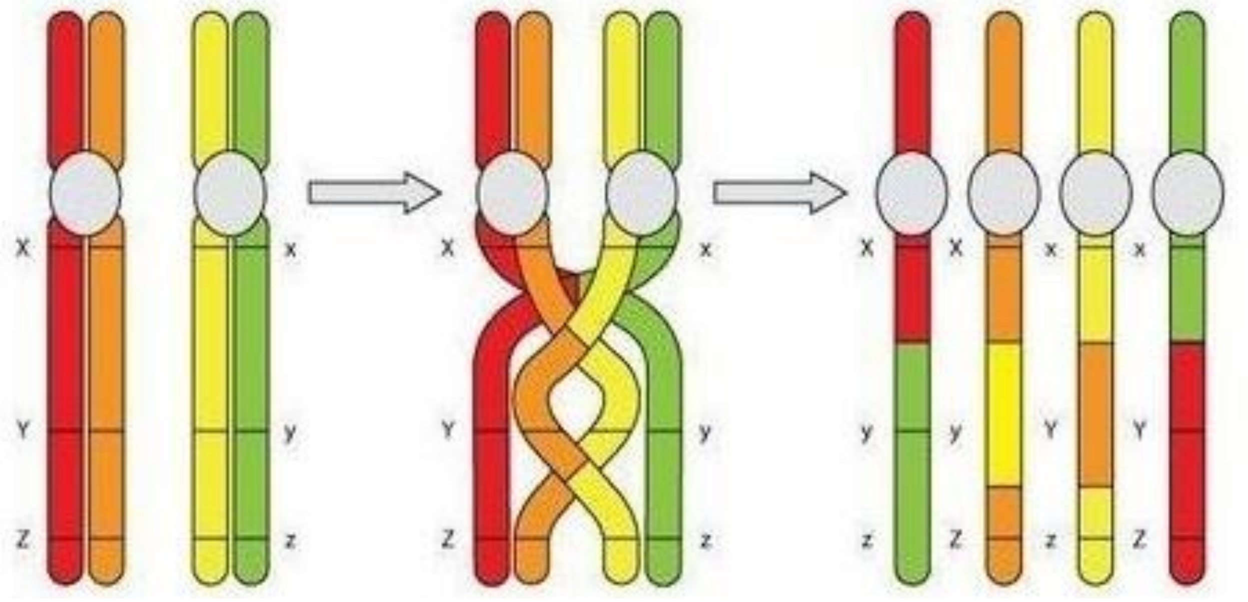
This individual with cri-du-chat syndrome is shown at two, four, nine, and 12 years of age.

Genetic Variation in Meiosis

Genetic variation is a result of the process of meiosis, which is a type of cellular division that produces gametes. Meiosis increases genetic variation in a number of ways, including:

- **Recombination**: Homologous chromosomes pair up and exchange genes at points called chiasma. This process creates new combinations of genetic information.
- **Independent assortment**: Chromosomes move randomly to opposite poles during meiosis, which reshuffles genes into unique combinations.
- **Random orientation**: During meiosis I, homologous pairs separate into two equal groups, but which chromosome goes into which group is determined randomly.

Genetic Variation in Meiosis

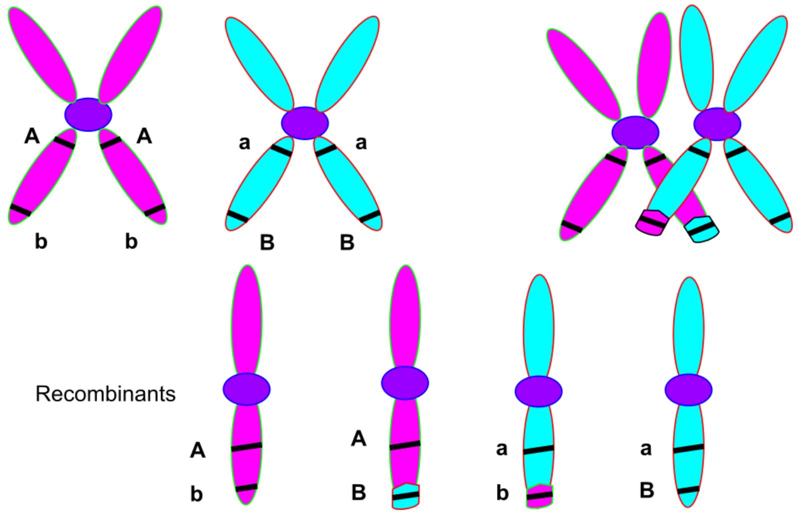


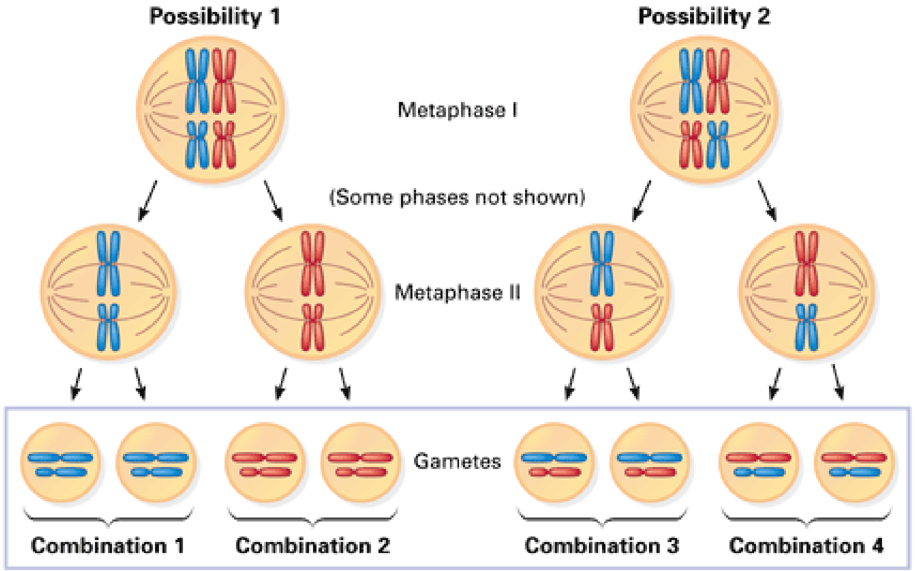
The gametes produced in meiosis aren't genetically identical to the starting cell, and they also aren't identical to one another. As an example, consider the meiosis II diagram above, which shows the end products of meiosis for a simple cell with a diploid number of **2*n* = 4** chromosomes. The four gametes produced at the end of meiosis II are all slightly different, each with a unique combination of the genetic material present in the starting cell.

As it turns out, there are many more potential gamete types than just the four shown in the diagram, even for a simple cell with only four chromosomes. This diversity of possible gametes reflects two factors: crossing over and the random orientation of homologue pairs during metaphase of meiosis I.

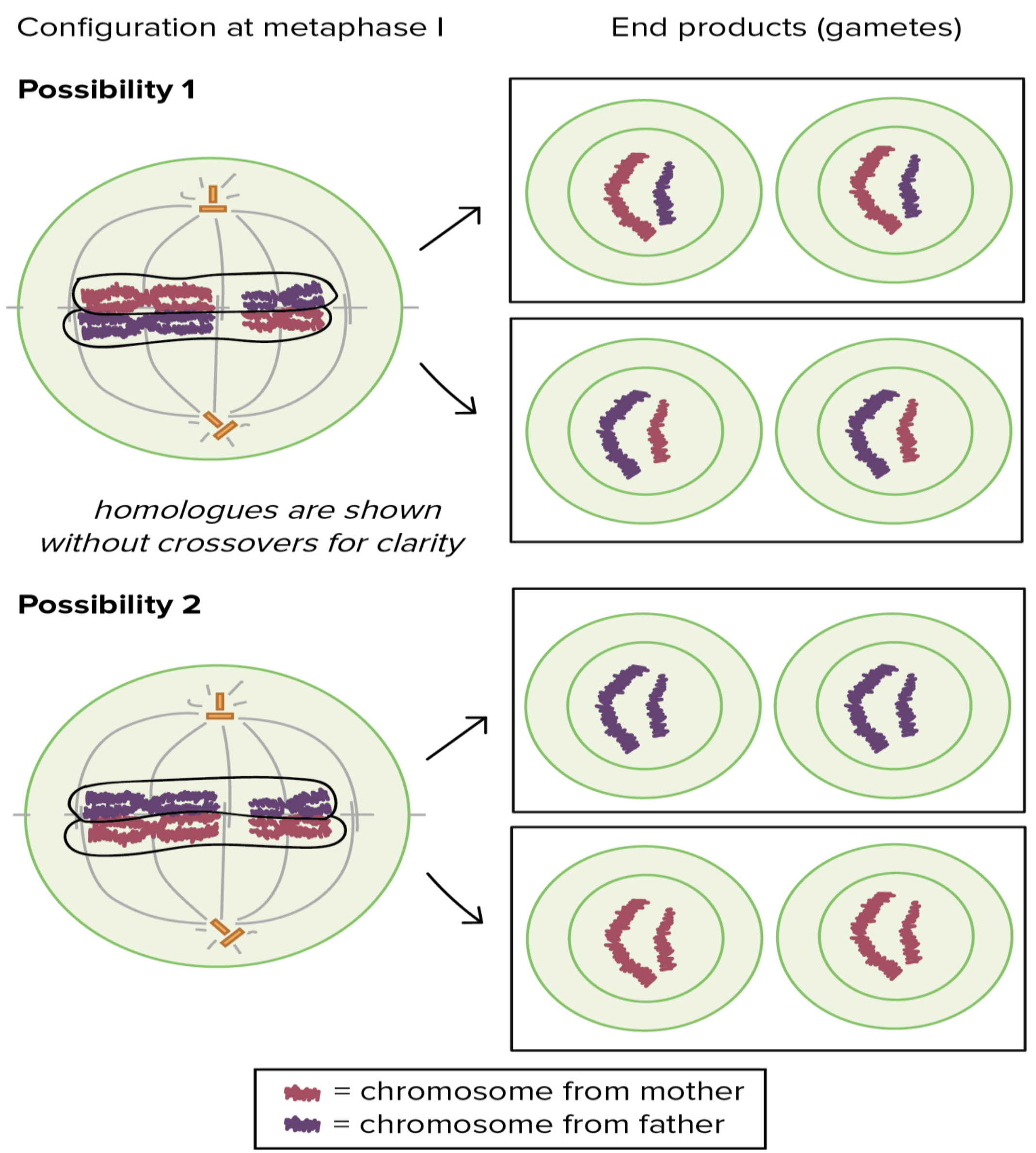
Genetic Variation in Meiosis

- **Crossing over.** The points where homologues cross over and exchange genetic material are chosen more or less at random, and they will be different in each cell that goes through meiosis. If meiosis happens many times, as it does in human ovaries and testes, crossovers will happen at many different points. This repetition produces a wide variety of recombinant chromosomes, chromosomes where fragments of DNA have been exchanged between homologues.
- **Random orientation of homologue pairs.** The random orientation of homologue pairs during metaphase of meiosis I is another important source of gamete diversity.

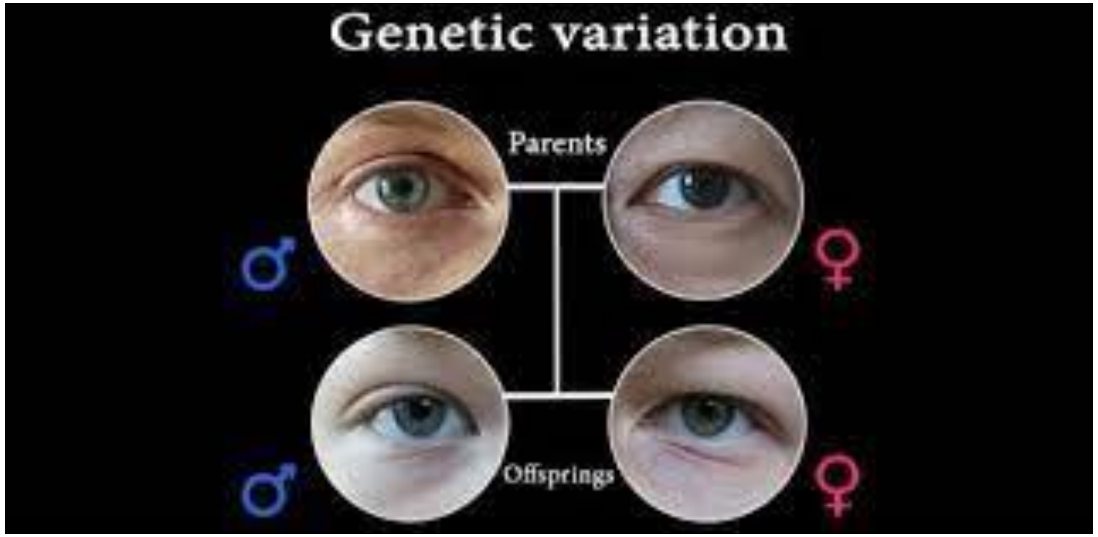




Genetic Variation in Meiosis



Genetic Variation in Meiosis



- A homologous pair consists of one homologue from your dad and one from your mom, and you have 23 pairs of homologous chromosomes all together, counting the X and Y as homologous for this purpose. During meiosis I, the homologous pairs will separate to form two equal groups, but it's not usually the case that all the paternal—dad—chromosomes will go into one group and all the maternal—mom—chromosomes into the other.
- Instead, each pair of homologues will effectively flip a coin to decide which chromosome goes into which group. In a cell with just two pairs of homologous chromosomes, like the one at right, random metaphase orientation allows for $2^2 = 4$ different types of possible gametes. In a human cell, the same mechanism allows for $2^{23} = 8,388,608$ different types of possible gametes. And that's not even considering crossovers!

Genetic Variation in Meiosis

- Given those kinds of numbers, it's very unlikely that any two sperm or egg cells made by a person will be the same. It's even more unlikely that you and your sister or brother will be genetically identical, unless you happen to be identical twins, thanks to the process of fertilization (in which a unique egg from Mom combines with a unique sperm from Dad, making a zygote whose genotype is well beyond one-in-a-trillion!).
- Meiosis and fertilization create genetic variation by making new combinations of gene variants (alleles). In some cases, these new combinations may make an organism more or less fit (able to survive and reproduce), thus providing the raw material for natural selection. Genetic variation is important in allowing a population to adapt via natural selection and thus survive in the long term.

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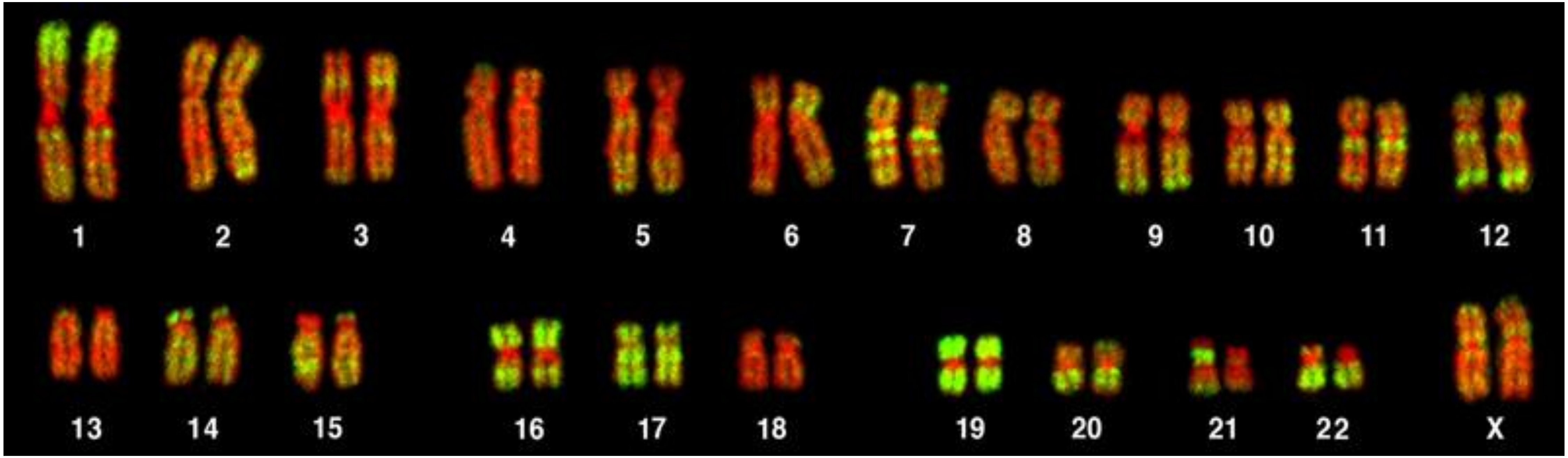
Chromosomal Karyotyping



- The isolation and microscopic observation of chromosomes forms the basis of cytogenetics and is the primary method by which clinicians detect chromosomal abnormalities in humans.
- A **karyotype** is the number and appearance of chromosomes, and includes their length, banding pattern, and centromere position. To obtain a view of an individual's karyotype, cytologists photograph the chromosomes and then cut and paste each chromosome into a chart, or **karyogram**, also known as an ideogram. The simplest use of a karyotype (or its karyogram image) is to identify abnormal chromosomal numbers.

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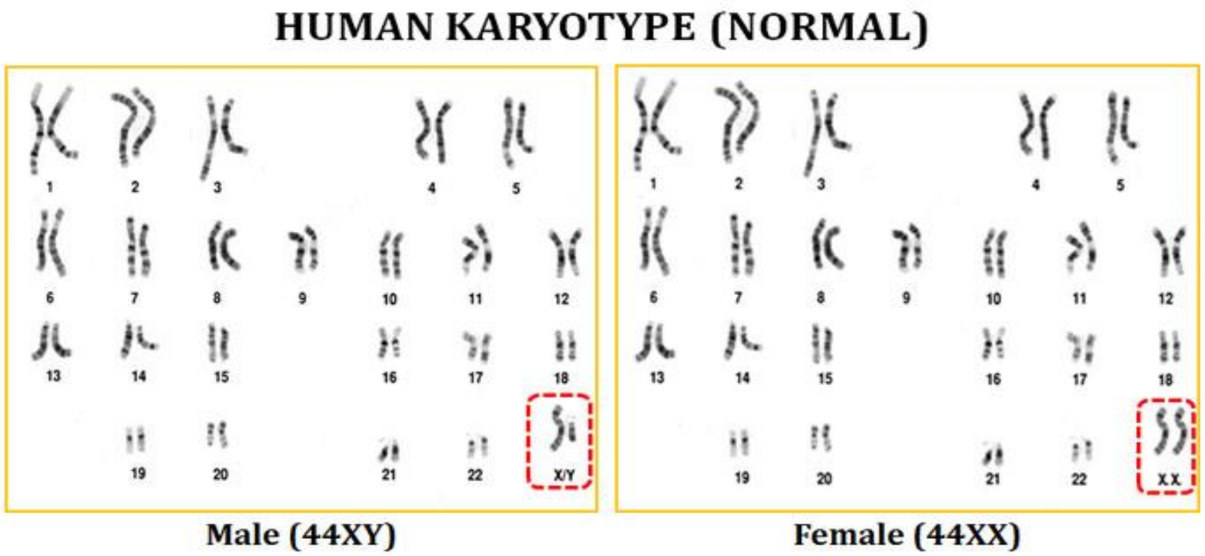
Karyotypes



- In a given species, chromosomes can be identified by their number, size, centromere position, and banding pattern. In a human karyotype, **autosomes** or “body chromosomes” (all of the non–sex chromosomes) are generally organized in approximate order of size from largest (chromosome 1) to smallest (chromosome 22). The X and Y chromosomes are not autosomes. However, chromosome 21 is actually shorter than chromosome 22. This was discovered after the naming of Down syndrome as trisomy 21, reflecting how this disease results from possessing one extra chromosome 21 (three total). Not wanting to change the name of this important disease, chromosome 21 retained its numbering, despite describing the shortest set of chromosomes. The chromosome “arms” projecting from either end of the centromere may be designated as short or long, depending on their relative lengths. The short arm is abbreviated *p* (for “petite”), whereas the long arm is abbreviated *q* (because it follows “p” alphabetically). Each arm is further subdivided and denoted by a number. Using this naming system, locations on chromosomes can be described consistently in the scientific literature.

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Common Disorders

- Of all of the chromosomal disorders, abnormalities in chromosome number are the most obviously identifiable from a karyogram. Disorders of chromosome number include the duplication or loss of entire chromosomes, as well as changes in the number of complete sets of chromosomes. They are caused by **nondisjunction**, which occurs when pairs of homologous chromosomes or sister chromatids fail to separate during meiosis. Misaligned or incomplete synapsis, or a dysfunction of the spindle apparatus that facilitates chromosome migration, can cause nondisjunction. The risk of nondisjunction occurring increases with the age of the parents.
- Nondisjunction can occur during either meiosis I or II, with differing results. If homologous chromosomes fail to separate during meiosis I, the result is two gametes that lack that particular chromosome and two gametes with two copies of the chromosome. If sister chromatids fail to separate during meiosis II, the result is one gamete that lacks that chromosome, two normal gametes with one copy of the chromosome, and one gamete with two copies of the chromosome.

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